

15

Integral Transform Method

EXERCISES 15.1

Error Function

1. (a) The result follows by letting $\tau = u^2$ or $u = \sqrt{\tau}$ in $\operatorname{erf}(\sqrt{t}) = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} e^{-u^2} du$.

(b) Using $\mathcal{L}\{t^{-1/2}\} = \frac{\sqrt{\pi}}{s^{1/2}}$ and the first translation theorem, it follows from the convolution theorem that

$$\begin{aligned} \mathcal{L}\{\operatorname{erf}(\sqrt{t})\} &= \frac{1}{\sqrt{\pi}} \mathcal{L}\left\{\int_0^t \frac{e^{-\tau}}{\sqrt{\tau}} d\tau\right\} = \frac{1}{\sqrt{\pi}} \mathcal{L}\{1\} \mathcal{L}\{t^{-1/2}e^{-t}\} = \frac{1}{\sqrt{\pi}} \frac{1}{s} \mathcal{L}\{t^{-1/2}\} \Big|_{s \rightarrow s+1} \\ &= \frac{1}{\sqrt{\pi}} \frac{1}{s} \frac{\sqrt{\pi}}{\sqrt{s+1}} = \frac{1}{s\sqrt{s+1}}. \end{aligned}$$

2. Since $\operatorname{erfc}(\sqrt{t}) = 1 - \operatorname{erf}(\sqrt{t})$ we have

$$\mathcal{L}\{\operatorname{erfc}(\sqrt{t})\} = \mathcal{L}\{1\} - \mathcal{L}\{\operatorname{erf}(\sqrt{t})\} = \frac{1}{s} - \frac{1}{s\sqrt{s+1}} = \frac{1}{s} \left[1 - \frac{1}{\sqrt{s+1}}\right].$$

3. By the first translation theorem,

$$\mathcal{L}\{e^t \operatorname{erf}(\sqrt{t})\} = \mathcal{L}\{\operatorname{erf}(\sqrt{t})\} \Big|_{s \rightarrow s-1} = \frac{1}{s\sqrt{s+1}} \Big|_{s \rightarrow s-1} = \frac{1}{\sqrt{s}(s-1)}.$$

4. By the first translation theorem and the result of Problem 2,

$$\begin{aligned} \mathcal{L}\{e^t \operatorname{erfc}(\sqrt{t})\} &= \mathcal{L}\{\operatorname{erfc}(\sqrt{t})\} \Big|_{s \rightarrow s-1} = \left(\frac{1}{s} - \frac{1}{s\sqrt{s+1}}\right) \Big|_{s \rightarrow s-1} = \frac{1}{s-1} - \frac{1}{\sqrt{s}(s-1)} \\ &= \frac{\sqrt{s}-1}{\sqrt{s}(s-1)} = \frac{\sqrt{s}-1}{\sqrt{s}(\sqrt{s}+1)(\sqrt{s}-1)} = \frac{1}{\sqrt{s}(\sqrt{s}+1)}. \end{aligned}$$

15.1 Error Function

5. From entry 3 in Table 15.1 and the first translation theorem we have

$$\begin{aligned}
 \mathcal{L}\left\{e^{-Gt/C} \operatorname{erf}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right\} &= \mathcal{L}\left\{e^{-Gt/C} \left[1 - \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right]\right\} \\
 &= \mathcal{L}\left\{e^{-Gt/C}\right\} - \mathcal{L}\left\{e^{-Gt/C} \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right\} \\
 &= \frac{1}{s + G/C} - \frac{e^{-x\sqrt{RC}}\sqrt{s}}{s} \Big|_{s \rightarrow s+G/C} \\
 &= \frac{1}{s + G/C} - \frac{e^{-x\sqrt{RC}}\sqrt{s+G/C}}{s + G/C} = \frac{C}{Cs + G} \left(1 - e^{x\sqrt{RCs+RG}}\right).
 \end{aligned}$$

6. We first compute

$$\begin{aligned}
 \frac{\sinh a\sqrt{s}}{s \sinh \sqrt{s}} &= \frac{e^{a\sqrt{s}} - e^{-a\sqrt{s}}}{s(e^{\sqrt{s}} - e^{-\sqrt{s}})} = \frac{e^{(a-1)\sqrt{s}} - e^{-(a+1)\sqrt{s}}}{s(1 - e^{-2\sqrt{s}})} \\
 &= \frac{e^{(a-1)\sqrt{s}}}{s} \left[1 + e^{-2\sqrt{s}} + e^{-4\sqrt{s}} + \dots\right] - \frac{e^{-(a+1)\sqrt{s}}}{s} \left[1 + e^{-2\sqrt{s}} + e^{-4\sqrt{s}} + \dots\right] \\
 &= \left[\frac{e^{-(1-a)\sqrt{s}}}{s} + \frac{e^{-(3-a)\sqrt{s}}}{s} + \frac{e^{-(5-a)\sqrt{s}}}{s} + \dots\right] \\
 &\quad - \left[\frac{e^{-(1+a)\sqrt{s}}}{s} + \frac{e^{-(3+a)\sqrt{s}}}{s} + \frac{e^{-(5+a)\sqrt{s}}}{s} + \dots\right] \\
 &= \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+1-a)\sqrt{s}}}{s} - \frac{e^{-(2n+1+a)\sqrt{s}}}{s}\right].
 \end{aligned}$$

Then

$$\begin{aligned}
 \mathcal{L}^{-1}\left\{\frac{\sinh a\sqrt{s}}{s \sinh \sqrt{s}}\right\} &= \sum_{n=0}^{\infty} \left[\mathcal{L}^{-1}\left\{\frac{e^{-(2n+1-a)\sqrt{s}}}{s}\right\} - \mathcal{L}^{-1}\left\{-\frac{e^{-(2n+1+a)\sqrt{s}}}{s}\right\}\right] \\
 &= \sum_{n=0}^{\infty} \left[\operatorname{erfc}\left(\frac{2n+1-a}{2\sqrt{t}}\right) - \operatorname{erfc}\left(\frac{2n+1+a}{2\sqrt{t}}\right)\right] \\
 &= \sum_{n=0}^{\infty} \left(\left[1 - \operatorname{erf}\left(\frac{2n+1-a}{2\sqrt{t}}\right)\right] - \left[1 - \operatorname{erf}\left(\frac{2n+1+a}{2\sqrt{t}}\right)\right]\right) \\
 &= \sum_{n=0}^{\infty} \left[\operatorname{erf}\left(\frac{2n+1+a}{2\sqrt{t}}\right) - \operatorname{erf}\left(\frac{2n+1-a}{2\sqrt{t}}\right)\right].
 \end{aligned}$$

7. Taking the Laplace transform of both sides of the equation we obtain

$$\begin{aligned}
 \mathcal{L}\{y(t)\} &= \mathcal{L}\{1\} - \mathcal{L}\left\{\int_0^t \frac{y(\tau)}{\sqrt{t-\tau}} d\tau\right\} \\
 Y(s) &= \frac{1}{s} - Y(s) \frac{\sqrt{\pi}}{\sqrt{s}} \\
 \frac{\sqrt{s} + \sqrt{\pi}}{\sqrt{s}} Y(s) &= \frac{1}{s} \\
 Y(s) &= \frac{1}{\sqrt{s}(\sqrt{s} + \sqrt{\pi})}.
 \end{aligned}$$

Thus

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{\sqrt{s}(\sqrt{s} + \sqrt{\pi})} \right\} = e^{\pi t} \operatorname{erfc}(\sqrt{\pi t}). \quad \boxed{\text{By entry 5 in Table 15.1}}$$

8. Using entries 3 and 5 in Table 15.1, we have

$$\begin{aligned} & \mathcal{L} \left\{ -e^{ab} e^{b^2 t} \operatorname{erfc} \left(b\sqrt{t} + \frac{a}{2\sqrt{t}} \right) + \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} \right) \right\} \\ &= -\mathcal{L} \left\{ e^{ab} e^{b^2 t} \operatorname{erfc} \left(b\sqrt{t} + \frac{a}{2\sqrt{t}} \right) \right\} + \mathcal{L} \left\{ \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} \right) \right\} \\ &= -\frac{e^{-a\sqrt{s}}}{\sqrt{s}(\sqrt{s} + b)} + \frac{e^{-a\sqrt{s}}}{s} \\ &= e^{-a\sqrt{s}} \left[\frac{1}{s} - \frac{1}{\sqrt{s}(\sqrt{s} + b)} \right] = e^{-a\sqrt{s}} \left[\frac{1}{s} - \frac{\sqrt{s}}{s(\sqrt{s} + b)} \right] \\ &= e^{-a\sqrt{s}} \left[\frac{\sqrt{s} + b - \sqrt{s}}{s(\sqrt{s} + b)} \right] = \frac{be^{-a\sqrt{s}}}{s(\sqrt{s} + b)}. \end{aligned}$$

$$\begin{aligned} 9. \quad \int_a^b e^{-u^2} du &= \int_a^0 e^{-u^2} du + \int_0^b e^{-u^2} du = \int_0^b e^{-u^2} du - \int_0^a e^{-u^2} du \\ &= \frac{\sqrt{\pi}}{2} \operatorname{erf}(b) - \frac{\sqrt{\pi}}{2} \operatorname{erf}(a) = \frac{\sqrt{\pi}}{2} [\operatorname{erf}(b) - \operatorname{erf}(a)] \end{aligned}$$

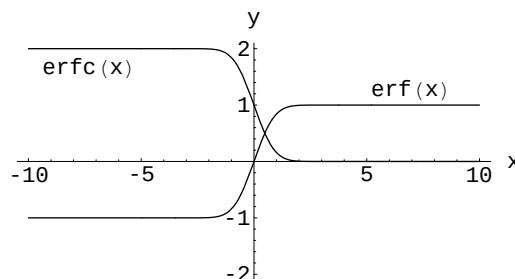
10. Since $f(x) = e^{-x^2}$ is an even function,

$$\int_{-a}^a e^{-u^2} du = 2 \int_0^a e^{-u^2} du.$$

Therefore,

$$\int_{-a}^a e^{-u^2} du = \sqrt{\pi} \operatorname{erf}(a).$$

11. The function $\operatorname{erf}(x)$ is symmetric with respect to the origin, while $\operatorname{erfc}(x)$ appears to be symmetric with respect to the point $(0, 1)$. From the graph it appears that $\lim_{x \rightarrow -\infty} \operatorname{erf}(x) = -1$ and $\lim_{x \rightarrow -\infty} \operatorname{erfc}(x) = 2$.



EXERCISES 15.2**Applications of the Laplace Transform**

1. The boundary-value problem is

$$\begin{aligned} a^2 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0, \\ u(0, t) &= 0, \quad u(L, t) = 0, \quad t > 0, \\ u(x, 0) &= A \sin \frac{\pi}{L} x, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0. \end{aligned}$$

Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - \left(\frac{s}{a}\right)^2 U = -\frac{s}{a^2} A \sin \frac{\pi}{L} x.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 \cosh \frac{s}{a} x + c_2 \sinh \frac{s}{a} x + \frac{As}{s^2 + a^2 \pi^2 / L^2} \sin \frac{\pi}{L} x.$$

The transformed boundary conditions, $U(0, s) = 0$, $U(L, s) = 0$ give in turn $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{As}{s^2 + a^2 \pi^2 / L^2} \sin \frac{\pi}{L} x$$

and

$$u(x, t) = A \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + a^2 \pi^2 / L^2} \right\} \sin \frac{\pi}{L} x = A \cos \frac{a\pi}{L} t \sin \frac{\pi}{L} x.$$

2. The transformed equation is

$$\frac{d^2 U}{dx^2} - s^2 U = -2 \sin \pi x - 4 \sin 3\pi x$$

and so

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{2}{s^2 + \pi^2} \sin \pi x + \frac{4}{s^2 + 9\pi^2} \sin 3\pi x.$$

The transformed boundary conditions, $U(0, s) = 0$ and $U(1, s) = 0$ give $c_1 = 0$ and $c_2 = 0$. Thus

$$U(x, s) = \frac{2}{s^2 + \pi^2} \sin \pi x + \frac{4}{s^2 + 9\pi^2} \sin 3\pi x$$

and

$$\begin{aligned} u(x, t) &= 2 \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + \pi^2} \right\} \sin \pi x + 4 \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9\pi^2} \right\} \sin 3\pi x \\ &= \frac{2}{\pi} \sin \pi t \sin \pi x + \frac{4}{3\pi} \sin 3\pi t \sin 3\pi x. \end{aligned}$$

3. The solution of

$$a^2 \frac{d^2 U}{dx^2} - s^2 U = 0$$

is in this case

$$U(x, s) = c_1 e^{-(x/a)s} + c_2 e^{(x/a)s}.$$

Since $\lim_{x \rightarrow \infty} u(x, t) = 0$ we have $\lim_{x \rightarrow \infty} U(x, s) = 0$. Thus $c_2 = 0$ and

$$U(x, s) = c_1 e^{-(x/a)s}.$$

If $\mathcal{L}\{u(0, t)\} = \mathcal{L}\{f(t)\} = F(s)$ then $U(0, s) = F(s)$. From this we have $c_1 = F(s)$ and

$$U(x, s) = F(s) e^{-(x/a)s}.$$

Hence, by the second translation theorem,

$$u(x, t) = f\left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a}\right).$$

4. Expressing $f(t)$ in the form $(\sin \pi t)[1 - \mathcal{U}(t - 1)]$ and using the result of Problem 3 we find

$$\begin{aligned} u(x, t) &= f\left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a}\right) \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[1 - \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \mathcal{U}\left(t - \frac{x}{a}\right) \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[\mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[\mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \end{aligned}$$

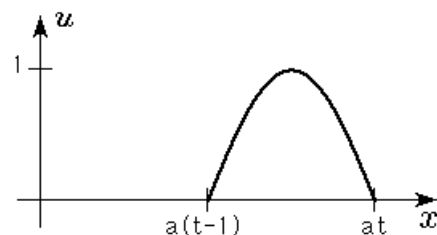
Now

$$\begin{aligned} \mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a} - 1\right) &= \begin{cases} 0, & 0 \leq t < x/a \\ 1, & x/a \leq t \leq x/a + 1 \\ 0, & t > x/a + 1 \end{cases} \\ &= \begin{cases} 0, & x < a(t - 1) \text{ or } x > at \\ 1, & a(t - 1) \leq x \leq at \end{cases} \end{aligned}$$

so

$$u(x, t) = \begin{cases} 0, & x < a(t - 1) \text{ or } x > at \\ \sin \pi(t - x/a), & a(t - 1) \leq x \leq at. \end{cases}$$

The graph is shown for $t > 1$.



5. We use

$$U(x, s) = c_1 e^{-(x/a)s} - \frac{g}{s^3}.$$

Now

$$\mathcal{L}\{u(0, t)\} = U(0, s) = \frac{A\omega}{s^2 + \omega^2}$$

and so

$$U(0, s) = c_1 - \frac{g}{s^3} = \frac{A\omega}{s^2 + \omega^2} \quad \text{or} \quad c_1 = \frac{g}{s^3} + \frac{A\omega}{s^2 + \omega^2}.$$

Therefore

$$U(x, s) = \frac{A\omega}{s^2 + \omega^2} e^{-(x/a)s} + \frac{g}{s^3} e^{-(x/a)s} - \frac{g}{s^3}$$

and

$$\begin{aligned} u(x, t) &= A \mathcal{L}^{-1} \left\{ \frac{\omega e^{-(x/a)s}}{s^2 + \omega^2} \right\} + g \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s^3} \right\} - g \mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} \\ &= A \sin \omega \left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a}\right) + \frac{1}{2} g \left(t - \frac{x}{a}\right)^2 \mathcal{U}\left(t - \frac{x}{a}\right) - \frac{1}{2} g t^2. \end{aligned}$$

15.2 Applications of the Laplace Transform

6. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - s^2 U = -\frac{\omega}{s^2 + \omega^2} \sin \pi x.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{\omega}{(s^2 + \pi^2)(s^2 + \omega^2)} \sin \pi x.$$

The transformed boundary conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{\omega}{(s^2 + \pi^2)(s^2 + \omega^2)} \sin \pi x$$

and

$$\begin{aligned} u(x, t) &= \omega \sin \pi x \mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \pi^2)(s^2 + \omega^2)} \right\} \\ &= \frac{\omega}{\omega^2 - \pi^2} \sin \pi x \mathcal{L}^{-1} \left\{ \frac{1}{\pi} \frac{\pi}{s^2 + \pi^2} - \frac{1}{\omega} \frac{\omega}{s^2 + \omega^2} \right\} \\ &= \frac{\omega}{\pi(\omega^2 - \pi^2)} \sin \pi t \sin \pi x - \frac{1}{\omega^2 - \pi^2} \sin \omega t \sin \pi x. \end{aligned}$$

7. We use

$$U(x, s) = c_1 \cosh \frac{s}{a} x + c_2 \sinh \frac{s}{a} x.$$

Now $U(0, s) = 0$ implies $c_1 = 0$, so $U(x, s) = c_2 \sinh(s/a)x$. The condition $E dU/dx \big|_{x=L} = F_0$ then yields $c_2 = F_0 a / Es \cosh(s/a)L$ and so

$$\begin{aligned} U(x, s) &= \frac{aF_0}{Es} \frac{\sinh(s/a)x}{\cosh(s/a)L} = \frac{aF_0}{Es} \frac{e^{(s/a)x} - e^{-(s/a)x}}{e^{(s/a)L} + e^{-(s/a)L}} \\ &= \frac{aF_0}{Es} \frac{e^{(s/a)(x-L)} - e^{-(s/a)(x+L)}}{1 + e^{-2sL/a}} \\ &= \frac{aF_0}{E} \left[\frac{e^{-(s/a)(L-x)}}{s} - \frac{e^{-(s/a)(3L-x)}}{s} + \frac{e^{-(s/a)(5L-x)}}{s} - \dots \right] \\ &\quad - \frac{aF_0}{E} \left[\frac{e^{-(s/a)(L+x)}}{s} - \frac{e^{-(s/a)(3L+x)}}{s} + \frac{e^{-(s/a)(5L+x)}}{s} - \dots \right] \\ &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\frac{e^{-(s/a)(2nL+L-x)}}{s} - \frac{e^{-(s/a)(2nL+L+x)}}{s} \right] \end{aligned}$$

and

$$\begin{aligned} u(x, t) &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(s/a)(2nL+L-x)}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(s/a)(2nL+L+x)}}{s} \right\} \right] \\ &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\left(t - \frac{2nL+L-x}{a} \right) \mathcal{U} \left(t - \frac{2nL+L-x}{a} \right) \right. \\ &\quad \left. - \left(t - \frac{2nL+L+x}{a} \right) \mathcal{U} \left(t - \frac{2nL+L+x}{a} \right) \right]. \end{aligned}$$

8. We use

$$U(x, s) = c_1 e^{-(x/a)s} + c_2 e^{(x/a)s} - \frac{v_0}{s^2}.$$

Now $\lim_{x \rightarrow \infty} dU/dx = 0$ implies $c_2 = 0$, and $U(0, s) = 0$ then gives $c_1 = v_0/s^2$. Hence

$$U(x, s) = \frac{v_0}{s^2} e^{-(x/a)s} - \frac{v_0}{s^2}$$

and

$$u(x, t) = v_0 \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) - v_0 t.$$

9. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - s^2 U = -sxe^{-x}.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-sx} + c_2 e^{sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x}.$$

The transformed boundary conditions $\lim_{x \rightarrow \infty} U(x, s) = 0$ and $U(0, s) = 0$ give, in turn, $c_2 = 0$ and $c_1 = 2s/(s^2 - 1)^2$. Therefore

$$U(x, s) = \frac{2s}{(s^2 - 1)^2} e^{-sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x}.$$

From entries (13) and (26) in the Table of Laplace transforms we obtain

$$\begin{aligned} u(x, t) &= \mathcal{L}^{-1} \left\{ \frac{2s}{(s^2 - 1)^2} e^{-sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x} \right\} \\ &= 2(t - x) \sinh(t - x) \mathcal{U}(t - x) - t e^{-x} \sinh t + x e^{-x} \cosh t. \end{aligned}$$

10. We use

$$U(x, s) = c_1 e^{-xs} + c_2 e^{xs} + \frac{s}{s^2 - 1} e^{-x}.$$

Now $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-xs} + \frac{s}{s^2 - 1} e^{-x}.$$

Finally, $U(0, s) = 1/s$ gives $c_1 = 1/s - s/(s^2 - 1)$. Thus

$$U(x, s) = \frac{1}{s} - \frac{s}{s^2 - 1} e^{-xs} + \frac{s}{s^2 - 1} e^{-x}$$

and

$$\begin{aligned} u(x, t) &= -\mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 1} e^{-(x/a)s} \right\} + \mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 1} \right\} e^{-x} \\ &= -\cosh \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) + e^{-x} \cosh t. \end{aligned}$$

11. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_1}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = u_1$ implies $\lim_{x \rightarrow \infty} U(x, s) = u_1/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_1}{s}.$$

From $U(0, s) = u_0/s$ we obtain $c_1 = (u_0 - u_1)/s$. Thus

$$U(x, s) = (u_0 - u_1) \frac{e^{-\sqrt{s}x}}{s} + \frac{u_1}{s}$$

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and

$$u(x, t) = (u_0 - u_1) \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s} \right\} + u_1 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = (u_0 - u_1) \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) + u_1.$$

12. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_1 x}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t)/x = u_1$ implies $\lim_{x \rightarrow \infty} U(x, s)/x = u_1/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_1 x}{s}.$$

From $U(0, s) = u_0/s$ we obtain $c_1 = u_0/s$. Hence

$$U(x, s) = u_0 \frac{e^{-\sqrt{s}x}}{s} + \frac{u_1 x}{s}$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s} \right\} + u_1 x \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) + u_1 x.$$

13. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_0}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = u_0$ implies $\lim_{x \rightarrow \infty} U(x, s) = u_0/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_0}{s}.$$

The transform of the remaining boundary conditions gives

$$\left. \frac{dU}{dx} \right|_{x=0} = U(0, s).$$

This condition yields $c_1 = -u_0/s(\sqrt{s} + 1)$. Thus

$$U(x, s) = -u_0 \frac{e^{-\sqrt{s}x}}{s(\sqrt{s} + 1)} + \frac{u_0}{s}$$

and

$$\begin{aligned} u(x, t) &= -u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} \\ &= u_0 e^{x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{x}{2\sqrt{t}} \right) - u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) + u_0 \end{aligned} \quad \boxed{\text{By entry (6) in Table 15.1}}$$

14. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Hence

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The remaining boundary condition transforms into

$$\left. \frac{dU}{dx} \right|_{x=0} = U(0, s) - \frac{50}{s}.$$

This condition gives $c_1 = 50/s(\sqrt{s} + 1)$. Therefore

$$U(x, s) = 50 \frac{e^{-\sqrt{s}x}}{s(\sqrt{s} + 1)}$$

and

$$u(x, t) = 50 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} = -50e^{x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{x}{2\sqrt{t}} \right) + 50 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right).$$

15. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Hence

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The transform of $u(0, t) = f(t)$ is $U(0, s) = F(s)$. Therefore

$$U(x, s) = F(s) e^{-\sqrt{s}x}$$

and

$$u(x, t) = \mathcal{L}^{-1} \left\{ F(s) e^{-x\sqrt{s}} \right\} = \frac{x}{2\sqrt{\pi}} \int_0^t \frac{f(t-\tau) e^{-x^2/4\tau}}{\tau^{3/2}} d\tau.$$

16. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then $U(x, s) = c_1 e^{-\sqrt{s}x}$.

The transform of the remaining boundary condition gives

$$\left. \frac{dU}{dx} \right|_{x=0} = -F(s)$$

where $F(s) = \mathcal{L}\{f(t)\}$. This condition yields $c_1 = F(s)/\sqrt{s}$. Thus

$$U(x, s) = F(s) \frac{e^{-\sqrt{s}x}}{\sqrt{s}}.$$

Using the Table of Laplace transforms and the convolution theorem we obtain

$$u(x, t) = \mathcal{L}^{-1} \left\{ F(s) \cdot \frac{e^{-\sqrt{s}x}}{\sqrt{s}} \right\} = \frac{1}{\sqrt{\pi}} \int_0^t f(\tau) \frac{e^{-x^2/4(t-\tau)}}{\sqrt{t-\tau}} d\tau.$$

17. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - sU = -60.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{60}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 60$ implies $\lim_{x \rightarrow \infty} U(x, s) = 60/s$, so we define $c_2 = 0$. The transform of the remaining boundary condition gives

$$U(0, s) = \frac{60}{s} + \frac{40}{s} e^{-2s}.$$

This condition yields $c_1 = \frac{40}{s} e^{-2s}$. Thus

$$U(x, s) = \frac{60}{s} + 40e^{-2s} \frac{e^{-\sqrt{s}x}}{s}.$$

Using the Table of Laplace transforms and the second translation theorem we obtain

$$u(x, t) = \mathcal{L}^{-1} \left\{ \frac{60}{s} + 40e^{-2s} \frac{e^{-\sqrt{s}x}}{s} \right\} = 60 + 40 \operatorname{erfc} \left(\frac{x}{2\sqrt{t-2}} \right) \mathcal{U}(t-2).$$

15.2 Applications of the Laplace Transform

18. The solution of the transformed equation

$$\frac{d^2U}{dx^2} - sU = -100$$

by undetermined coefficients is

$$U(x, s) = c_1 e^{\sqrt{s}x} + c_2 e^{-\sqrt{s}x} + \frac{100}{s}.$$

From the fact that $\lim_{x \rightarrow \infty} U(x, s) = 100/s$ we see that $c_1 = 0$. Thus

$$U(x, s) = c_2 e^{-\sqrt{s}x} + \frac{100}{s}. \quad (1)$$

Now the transform of the boundary condition at $x = 0$ is

$$U(0, s) = 20 \left[\frac{1}{s} - \frac{1}{s} e^{-s} \right].$$

It follows from (1) that

$$\frac{20}{s} - \frac{20}{s} e^{-s} = c_2 + \frac{100}{s} \quad \text{or} \quad c_2 = -\frac{80}{s} - \frac{20}{s} e^{-s}$$

and so

$$\begin{aligned} U(x, s) &= \left(-\frac{80}{s} - \frac{20}{s} e^{-s} \right) e^{-\sqrt{s}x} + \frac{100}{s} \\ &= \frac{100}{s} - \frac{80}{s} e^{-\sqrt{s}x} - \frac{20}{s} e^{-\sqrt{s}x} e^{-s}. \end{aligned}$$

Thus

$$\begin{aligned} u(x, t) &= 100 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - 80 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} \right\} - 20 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} e^{-s} \right\} \\ &= 100 - 80 \operatorname{erfc}(x/2\sqrt{t}) - 20 \operatorname{erfc}(x/2\sqrt{t-1}) \mathcal{U}(t-1). \end{aligned}$$

19. Transforming the partial differential equation gives

$$\frac{d^2U}{dx^2} - sU = 0$$

and so

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow -\infty} u(x, t) = 0$ implies $\lim_{x \rightarrow -\infty} U(x, s) = 0$, so we define $c_1 = 0$. The transform of the remaining boundary condition gives

$$\left. \frac{dU}{dx} \right|_{x=1} = \frac{100}{s} - U(1, s).$$

This condition yields

$$c_2 \sqrt{s} e^{\sqrt{s}} = \frac{100}{s} - c_2 e^{\sqrt{s}}$$

from which it follows that

$$c_2 = \frac{100}{s(\sqrt{s} + 1)} e^{-\sqrt{s}}.$$

Thus

$$U(x, s) = 100 \frac{e^{-(1-x)\sqrt{s}}}{s(\sqrt{s} + 1)}.$$

Using the Table of Laplace transforms we obtain

$$u(x, t) = 100 \mathcal{L}^{-1} \left\{ \frac{e^{-(1-x)\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} = 100 \left[-e^{1-x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{1-x}{\sqrt{t}} \right) + \operatorname{erfc} \left(\frac{1-x}{2\sqrt{t}} \right) \right].$$

20. Transforming the partial differential equation gives

$$k \frac{d^2 U}{dx^2} - sU = -\frac{r}{s}.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x} + \frac{r}{s^2}.$$

The condition $\lim_{x \rightarrow \infty} \partial u / \partial x = 0$ implies $\lim_{x \rightarrow \infty} dU/dx = 0$, so we define $c_2 = 0$. The transform of the remaining boundary condition gives $U(0, s) = 0$. This condition yields $c_1 = -r/s^2$. Thus

$$U(x, s) = r \left[\frac{1}{s^2} - \frac{e^{-\sqrt{s/k}x}}{s^2} \right].$$

Using the Table of Laplace transforms and the convolution theorem we obtain

$$u(x, t) = r \mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{1}{s} \cdot \frac{e^{-\sqrt{s/k}x}}{s} \right\} = rt - r \int_0^t \operatorname{erfc} \left(\frac{x}{2\sqrt{k\tau}} \right) d\tau.$$

21. The solution of

$$\frac{d^2 U}{dx^2} - sU = -u_0 - u_0 \sin \frac{\pi}{L}x$$

is

$$U(x, s) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x) + \frac{u_0}{s} + \frac{u_0}{s + \pi^2/L^2} \sin \frac{\pi}{L}x.$$

The transformed boundary conditions $U(0, s) = u_0/s$ and $U(L, s) = u_0/s$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{u_0}{s} + \frac{u_0}{s + \pi^2/L^2} \sin \frac{\pi}{L}x$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s + \pi^2/L^2} \right\} \sin \frac{\pi}{L}x = u_0 + u_0 e^{-\pi^2 t/L^2} \sin \frac{\pi}{L}x.$$

22. The transform of the partial differential equation is

$$k \frac{d^2 U}{dx^2} - hU + h \frac{u_m}{s} = sU - u_0$$

or

$$k \frac{d^2 U}{dx^2} - (h + s)U = -h \frac{u_m}{s} - u_0.$$

By undetermined coefficients we find

$$U(x, s) = c_1 e^{\sqrt{(h+s)/k}x} + c_2 e^{-\sqrt{(h+s)/k}x} + \frac{hu_m + u_0 s}{s(s + h)}.$$

The transformed boundary conditions are $U'(0, s) = 0$ and $U'(L, s) = 0$. These conditions imply $c_1 = 0$ and $c_2 = 0$. By partial fractions we then get

$$U(x, s) = \frac{hu_m + u_0 s}{s(s + h)} = \frac{u_m}{s} - \frac{u_m}{s + h} + \frac{u_0}{s + h}.$$

Therefore,

$$u(x, t) = u_m \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - u_m \mathcal{L}^{-1} \left\{ \frac{1}{s + h} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s + h} \right\} = u_m - u_m e^{-ht} + u_0 e^{-ht}.$$

15.2 Applications of the Laplace Transform

23. We use

$$U(x, s) = c_1 \cosh \sqrt{\frac{s}{k}} x + c_2 \sinh \sqrt{\frac{s}{k}} x + \frac{u_0}{s}.$$

The transformed boundary conditions $dU/dx|_{x=0} = 0$ and $U(1, s) = 0$ give, in turn, $c_2 = 0$ and $c_1 = -u_0/s \cosh \sqrt{s/k}$. Therefore

$$\begin{aligned} U(x, s) &= \frac{u_0}{s} - \frac{u_0 \cosh \sqrt{s/k} x}{s \cosh \sqrt{s/k}} = \frac{u_0}{s} - u_0 \frac{e^{\sqrt{s/k} x} + e^{-\sqrt{s/k} x}}{s(e^{\sqrt{s/k}} + e^{-\sqrt{s/k}})} \\ &= \frac{u_0}{s} - u_0 \frac{e^{\sqrt{s/k}(x-1)} + e^{-\sqrt{s/k}(x+1)}}{s(1 + e^{-2\sqrt{s/k}})} \\ &= \frac{u_0}{s} - u_0 \left[\frac{e^{-\sqrt{s/k}(1-x)}}{s} - \frac{e^{-\sqrt{s/k}(3-x)}}{s} + \frac{e^{-\sqrt{s/k}(5-x)}}{s} - \dots \right] \\ &\quad - u_0 \left[\frac{e^{-\sqrt{s/k}(1+x)}}{s} - \frac{e^{-\sqrt{s/k}(3+x)}}{s} + \frac{e^{-\sqrt{s/k}(5+x)}}{s} - \dots \right] \\ &= \frac{u_0}{s} - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\frac{e^{-(2n+1-x)\sqrt{s/k}}}{s} + \frac{e^{-(2n+1+x)\sqrt{s/k}}}{s} \right] \end{aligned}$$

and

$$\begin{aligned} u(x, t) &= u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(2n+1-x)\sqrt{s/k}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(2n+1+x)\sqrt{s/k}}}{s} \right\} \right] \\ &= u_0 - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{kt}} \right) - \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{kt}} \right) \right]. \end{aligned}$$

24. We use

$$c(x, s) = c_1 \cosh \sqrt{\frac{s}{D}} x + c_2 \sinh \sqrt{\frac{s}{D}} x.$$

The transform of the two boundary conditions are $c(0, s) = c_0/s$ and $c(1, s) = c_0/s$. From these conditions we obtain $c_1 = c_0/s$ and

$$c_2 = c_0(1 - \cosh \sqrt{s/D})/s \sinh \sqrt{s/D}.$$

Therefore

$$\begin{aligned} c(x, s) &= c_0 \left[\frac{\cosh \sqrt{s/D} x}{s} + \frac{(1 - \cosh \sqrt{s/D})}{s \sinh \sqrt{s/D}} \sinh \sqrt{s/D} x \right] \\ &= c_0 \left[\frac{\sinh \sqrt{s/D} (1-x)}{s \sinh \sqrt{s/D}} + \frac{\sinh \sqrt{s/D} x}{s \sinh \sqrt{s/D}} \right] \\ &= c_0 \left[\frac{e^{\sqrt{s/D}(1-x)} - e^{-\sqrt{s/D}(1-x)}}{s(e^{\sqrt{s/D}} - e^{-\sqrt{s/D}})} + \frac{e^{\sqrt{s/D} x} - e^{-\sqrt{s/D} x}}{s(e^{\sqrt{s/D}} - e^{-\sqrt{s/D}})} \right] \\ &= c_0 \left[\frac{e^{-\sqrt{s/D} x} - e^{-\sqrt{s/D}(2-x)}}{s(1 - e^{-2\sqrt{s/D}})} + \frac{e^{\sqrt{s/D}(x-1)} - e^{-\sqrt{s/D}(x+1)}}{s(1 - e^{-2\sqrt{s/D}})} \right] \end{aligned}$$

$$\begin{aligned}
 &= c_0 \frac{(e^{-\sqrt{s/D}x} - e^{-\sqrt{s/D}(2-x)})}{s} \left(1 + e^{-2\sqrt{s/D}} + e^{-4\sqrt{s/D}} + \dots\right) \\
 &\quad + c_0 \frac{(e^{\sqrt{s/D}(x-1)} - e^{-\sqrt{s/D}(x+1)})}{s} \left(1 + e^{-2\sqrt{s/D}} + e^{-4\sqrt{s/D}} + \dots\right) \\
 &= c_0 \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+x)\sqrt{s/D}}}{s} - \frac{e^{-(2n+2-x)\sqrt{s/D}}}{s} \right] \\
 &\quad + c_0 \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+1-x)\sqrt{s/D}}}{s} - \frac{e^{-(2n+1+x)\sqrt{s/D}}}{s} \right]
 \end{aligned}$$

and so

$$\begin{aligned}
 c(x, t) &= c_0 \sum_{n=0}^{\infty} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+2-x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} \right] \\
 &\quad + c_0 \sum_{n=0}^{\infty} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+1-x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+1+x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} \right] \\
 &= c_0 \sum_{n=0}^{\infty} \left[\operatorname{erfc} \left(\frac{2n+x}{2\sqrt{Dt}} \right) - \operatorname{erfc} \left(\frac{2n+2-x}{2\sqrt{Dt}} \right) \right] \\
 &\quad + c_0 \sum_{n=0}^{\infty} \left[\operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{Dt}} \right) - \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{Dt}} \right) \right].
 \end{aligned}$$

Now using $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$ we get

$$\begin{aligned}
 c(x, t) &= c_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+2-x}{2\sqrt{Dt}} \right) - \operatorname{erf} \left(\frac{2n+x}{2\sqrt{Dt}} \right) \right] \\
 &\quad + c_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+1+x}{2\sqrt{Dt}} \right) - \operatorname{erf} \left(\frac{2n+1-x}{2\sqrt{Dt}} \right) \right].
 \end{aligned}$$

25. We use

$$U(x, s) = c_1 e^{-\sqrt{RCs+RG}x} + c_2 e^{\sqrt{RCs+RG}x} + \frac{Cu_0}{Cs+G}.$$

The condition $\lim_{x \rightarrow \infty} \partial u / \partial x = 0$ implies $\lim_{x \rightarrow \infty} dU/dx = 0$, so we define $c_2 = 0$. Applying $U(0, s) = 0$ to

$$U(x, s) = c_1 e^{-\sqrt{RCs+RG}x} + \frac{Cu_0}{Cs+G}$$

gives $c_1 = -Cu_0/(Cs+G)$. Therefore

$$U(x, s) = -Cu_0 \frac{e^{-\sqrt{RCs+RG}x}}{Cs+G} + \frac{Cu_0}{Cs+G}$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s+G/C} \right\} - u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{RC}\sqrt{s+G/C}}}{s+G/C} \right\}$$

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$$\begin{aligned}
&= u_0 e^{-Gt/C} - u_0 e^{-Gt/C} \operatorname{erfc}\left(\frac{x\sqrt{RC}}{2\sqrt{t}}\right) \\
&= u_0 e^{-Gt/C} \left[1 - \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right] \\
&= u_0 e^{-Gt/C} \operatorname{erf}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right).
\end{aligned}$$

26. (a) We use

$$U(x, s) = c_1 e^{-(s/a)x} + c_2 e^{(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/v_0)x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we must define $c_2 = 0$. Consequently

$$U(x, s) = c_1 e^{-(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/v_0)x}.$$

The remaining boundary condition transforms into $U(0, s) = 0$. From this we find

$$c_1 = -v_0^2 F_0 / (a^2 - v_0^2)s^2.$$

Therefore, by the second translation theorem

$$U(x, s) = -\frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/v_0)x}$$

and

$$\begin{aligned}
u(x, t) &= \frac{v_0^2 F_0}{a^2 - v_0^2} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(x/v_0)s}}{s^2} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s^2} \right\} \right] \\
&= \frac{v_0^2 F_0}{a^2 - v_0^2} \left[\left(t - \frac{x}{v_0} \right) \mathcal{U} \left(t - \frac{x}{v_0} \right) - \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) \right].
\end{aligned}$$

(b) In the case when $v_0 = a$ the solution of the transformed equation is

$$U(x, s) = c_1 e^{-(s/a)x} + c_2 e^{(s/a)x} - \frac{F_0}{2as} x e^{-(s/a)x}.$$

The usual analysis then leads to $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = -\frac{F_0}{2as} x e^{-(s/a)x}$$

and

$$u(x, t) = -\frac{x F_0}{2a} \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s} \right\} = -\frac{x F_0}{2a} \mathcal{U} \left(t - \frac{x}{a} \right).$$

27. We use

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + c_2 e^{\sqrt{s+h}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we take $c_2 = 0$. Therefore

$$U(x, s) = c_1 e^{-\sqrt{s+h}x}.$$

The Laplace transform of $u(0, t) = u_0$ is $U(0, s) = u_0/s$ and so

$$U(x, s) = u_0 \frac{e^{-\sqrt{s+h}x}}{s}$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s+h}x}}{s} \right\} = u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} e^{-\sqrt{s+h}x} \right\}.$$

From the first translation theorem,

$$\mathcal{L}^{-1} \left\{ e^{-\sqrt{s+h}x} \right\} = e^{-ht} \mathcal{L}^{-1} \{ e^{-x\sqrt{s}} \} = e^{-ht} \frac{x}{2\sqrt{\pi t^3}} e^{-x^2/4t}.$$

Thus, from the convolution theorem we obtain

$$u(x, s) = \frac{u_0 x}{2\sqrt{\pi}} \int_0^t \frac{e^{-h\tau - x^2/4\tau}}{\tau^{3/2}} d\tau.$$

28. (a) We use

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x}.$$

Now $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s/k}x}.$$

Finally, from $U(0, s) = u_0/s$ we obtain $c_1 = u_0/s$. Thus

$$U(x, s) = u_0 \frac{e^{-\sqrt{s/k}x}}{s}$$

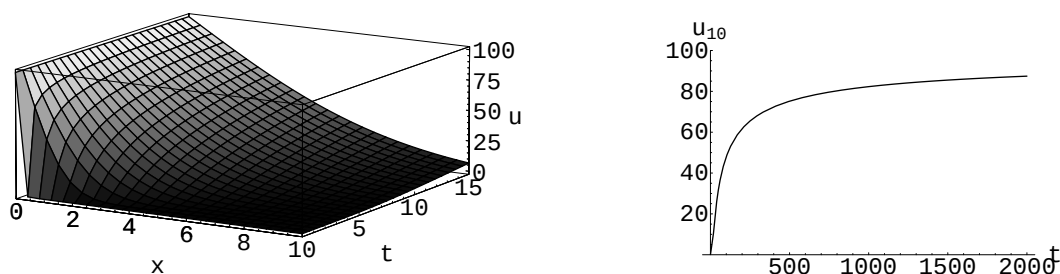
and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s/k}x}}{s} \right\} = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-(x/\sqrt{k})\sqrt{s}}}{s} \right\} = u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{kt}} \right).$$

Since $\operatorname{erfc}(0) = 1$,

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} u_0 \operatorname{erfc}(x/2\sqrt{kt}) = u_0.$$

(b)



29. (a) Transforming the partial differential equation and using the initial condition gives

$$k \frac{d^2 U}{dx^2} - sU = 0.$$

Since the domain of the variable x is an infinite interval we write the general solution of this differential equation as

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x}.$$

Transforming the boundary conditions gives $U'(0, s) = -A/s$ and $\lim_{x \rightarrow \infty} U(x, s) = 0$. Hence we find $c_2 = 0$ and $c_1 = A\sqrt{k}/s\sqrt{s}$. From

$$U(x, s) = A\sqrt{k} \frac{e^{-\sqrt{s/k}x}}{s\sqrt{s}}$$

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we see that

$$u(x, t) = A\sqrt{k} \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s/k}x}}{s\sqrt{s}} \right\}.$$

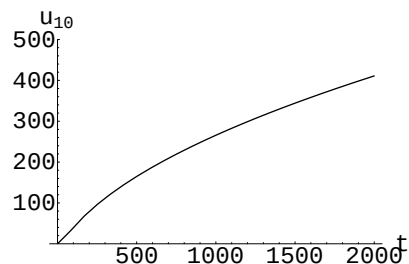
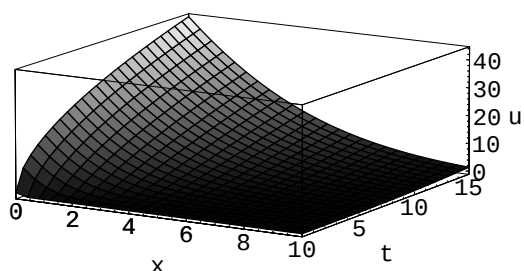
With the identification $a = x/\sqrt{k}$ it follows from the Table of Laplace transforms that

$$\begin{aligned} u(x, t) &= A\sqrt{k} \left\{ 2\sqrt{\frac{t}{\pi}} e^{-x^2/4kt} - \frac{x}{\sqrt{k}} \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) \right\} \\ &= 2A\sqrt{\frac{kt}{\pi}} e^{-x^2/4kt} - Ax \operatorname{erfc}\left(x/2\sqrt{kt}\right). \end{aligned}$$

Since $\operatorname{erfc}(0) = 1$,

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} \left(2A\sqrt{\frac{kt}{\pi}} e^{-x^2/4kt} - Ax \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) \right) = \infty.$$

(b)



30. (a) Letting $C(x, s) = \mathcal{L}\{c(x, t)\}$ we obtain

$$\frac{d^2 C}{dx^2} - \frac{s}{k} C = 0 \quad \text{subject to} \quad \left. \frac{dC}{dx} \right|_{x=0} = -A.$$

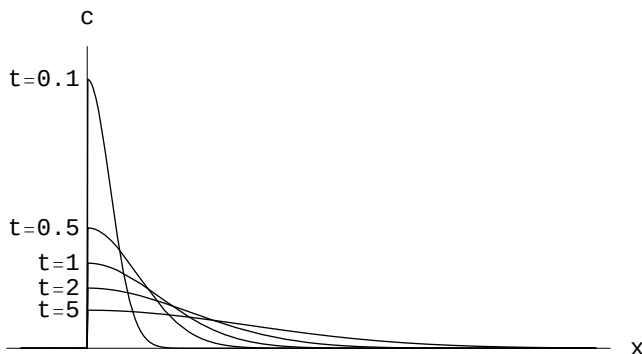
The solution of this initial-value problem is

$$C(x, s) = A\sqrt{k} \frac{e^{-(x/\sqrt{k})\sqrt{s}}}{\sqrt{s}},$$

so that

$$c(x, t) = A\sqrt{\frac{k}{\pi t}} e^{-x^2/4kt}.$$

(b)



(c) $\int_0^\infty c(x, t) dx = Ak \operatorname{erf}\left(\frac{x}{2\sqrt{kt}}\right) \Big|_0^\infty = Ak(1 - 0) = Ak$

EXERCISES 15.3

Fourier Integral

1. From formulas (5) and (6) in the text,

$$A(\alpha) = \int_{-1}^0 (-1) \cos \alpha x \, dx + \int_0^1 (2) \cos \alpha x \, dx = -\frac{\sin \alpha}{\alpha} + 2\frac{\sin \alpha}{\alpha} = \frac{\sin \alpha}{\alpha}$$

and

$$\begin{aligned} B(\alpha) &= \int_{-1}^0 (-1) \sin \alpha x \, dx + \int_0^1 (2) \sin \alpha x \, dx \\ &= \frac{1 - \cos \alpha}{\alpha} - 2\frac{\cos \alpha - 1}{\alpha} = \frac{3(1 - \cos \alpha)}{\alpha}. \end{aligned}$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{\sin \alpha \cos \alpha x + 3(1 - \cos \alpha) \sin \alpha x}{\alpha} d\alpha.$$

2. From formulas (5) and (6) in the text,

$$A(\alpha) = \int_\pi^{2\pi} 4 \cos \alpha x \, dx = 4 \frac{\sin 2\pi\alpha - \sin \pi\alpha}{\alpha}$$

and

$$B(\alpha) = \int_\pi^{2\pi} 4 \sin \alpha x \, dx = 4 \frac{\cos \pi\alpha - \cos 2\pi\alpha}{\alpha}.$$

Hence

$$\begin{aligned} f(x) &= \frac{4}{\pi} \int_0^\infty \frac{(\sin 2\pi\alpha - \sin \pi\alpha) \cos \alpha x + (\cos \pi\alpha - \cos 2\pi\alpha) \sin \alpha x}{\alpha} d\alpha \\ &= \frac{4}{\pi} \int_0^\infty \frac{\sin 2\pi\alpha \cos \alpha x - \cos 2\pi\alpha \sin \alpha x - \sin \pi\alpha \cos \alpha x + \cos \pi\alpha \sin \alpha x}{\alpha} d\alpha \\ &= \frac{4}{\pi} \int_0^\infty \frac{\sin \alpha(2\pi - x) - \sin \alpha(\pi - x)}{\alpha} d\alpha. \end{aligned}$$

3. From formulas (5) and (6) in the text,

$$\begin{aligned} A(\alpha) &= \int_0^3 x \cos \alpha x \, dx = \left. \frac{x \sin \alpha x}{\alpha} \right|_0^3 - \frac{1}{\alpha} \int_0^3 \sin \alpha x \, dx \\ &= \left. \frac{3 \sin 3\alpha}{\alpha} + \frac{\cos \alpha x}{\alpha^2} \right|_0^3 = \frac{3\alpha \sin 3\alpha + \cos 3\alpha - 1}{\alpha^2} \end{aligned}$$

and

$$\begin{aligned} B(\alpha) &= \int_0^3 x \sin \alpha x \, dx = -\left. \frac{x \cos \alpha x}{\alpha} \right|_0^3 + \frac{1}{\alpha} \int_0^3 \cos \alpha x \, dx \\ &= -\left. \frac{3 \cos 3\alpha}{\alpha} + \frac{\sin \alpha x}{\alpha^2} \right|_0^3 = \frac{\sin 3\alpha - 3\alpha \cos 3\alpha}{\alpha^2}. \end{aligned}$$

15.3 Fourier Integral

Hence

$$\begin{aligned}
 f(x) &= \frac{1}{\pi} \int_0^\infty \frac{(3\alpha \sin 3\alpha + \cos 3\alpha - 1) \cos \alpha x + (\sin 3\alpha - 3\alpha \cos 3\alpha) \sin \alpha x}{\alpha^2} d\alpha \\
 &= \frac{1}{\pi} \int_0^\infty \frac{3\alpha(\sin 3\alpha \cos \alpha x - \cos 3\alpha \sin \alpha x) + \cos 3\alpha \cos \alpha x + \sin 3\alpha \sin \alpha x - \cos \alpha x}{\alpha^2} d\alpha \\
 &= \frac{1}{\pi} \int_0^\infty \frac{3\alpha \sin \alpha(3-x) + \cos \alpha(3-x) - \cos \alpha x}{\alpha^2} d\alpha.
 \end{aligned}$$

4. From formulas (5) and (6) in the text,

$$\begin{aligned}
 A(\alpha) &= \int_{-\infty}^\infty f(x) \cos \alpha x dx \\
 &= \int_{-\infty}^0 0 \cdot \cos \alpha x dx + \int_0^\pi \sin x \cos \alpha x dx + \int_\pi^\infty 0 \cdot \cos \alpha x dx \\
 &= \frac{1}{2} \int_0^\pi [\sin(1+\alpha)x + \sin(1-\alpha)x] dx \\
 &= \frac{1}{2} \left[-\frac{\cos(1+\alpha)x}{1+\alpha} - \frac{\cos(1-\alpha)x}{1-\alpha} \right]_0^\pi \\
 &= -\frac{1}{2} \left[\frac{\cos(1+\alpha)\pi - 1}{1+\alpha} + \frac{\cos(1-\alpha)\pi - 1}{1-\alpha} \right] \\
 &= -\frac{1}{2} \left[\frac{\cos(1+\alpha)\pi - \alpha \cos(1+\alpha)\pi + \cos(1-\alpha)\pi + \alpha \cos(1-\alpha)\pi - 2}{1-\alpha^2} \right] \\
 &= \frac{1 + \cos \alpha\pi}{1 - \alpha^2},
 \end{aligned}$$

and

$$\begin{aligned}
 B(\alpha) &= \int_0^\pi \sin x \sin \alpha x dx = \frac{1}{2} \int_0^\pi [\cos(1-\alpha)x - \cos(1+\alpha)x] dx \\
 &= \frac{1}{2} \left[\frac{\sin(1-\alpha)\pi}{1-\alpha} - \frac{\sin(1+\alpha)\pi}{1+\alpha} \right] = \frac{\sin \alpha\pi}{1 - \alpha^2}.
 \end{aligned}$$

Hence

$$\begin{aligned}
 f(x) &= \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \cos \alpha x \cos \alpha\pi + \sin \alpha x \sin \alpha\pi}{1 - \alpha^2} d\alpha \\
 &= \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \cos \alpha(x - \pi)}{1 - \alpha^2} d\alpha.
 \end{aligned}$$

5. From formula (5) in the text,

$$A(\alpha) = \int_0^\infty e^{-x} \cos \alpha x dx.$$

Recall $\mathcal{L}\{\cos kt\} = s/(s^2 + k^2)$. If we set $s = 1$ and $k = \alpha$ we obtain

$$A(\alpha) = \frac{1}{1 + \alpha^2}.$$

Now

$$B(\alpha) = \int_0^\infty e^{-x} \sin \alpha x dx.$$

Recall $\mathcal{L}\{\sin kt\} = k/(s^2 + k^2)$. If we set $s = 1$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{\alpha}{1 + \alpha^2}.$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \alpha \sin \alpha x}{1 + \alpha^2} d\alpha.$$

6. From formulas (5) and (6) in the text,

$$\begin{aligned} A(\alpha) &= \int_{-1}^1 e^x \cos \alpha x dx \\ &= \frac{e(\cos \alpha + \alpha \sin \alpha) - e^{-1}(\cos \alpha - \alpha \sin \alpha)}{1 + \alpha^2} \\ &= \frac{2(\sinh 1) \cos \alpha - 2\alpha(\cosh 1) \sin \alpha}{1 + \alpha^2} \end{aligned}$$

and

$$\begin{aligned} B(\alpha) &= \int_{-1}^1 e^x \sin \alpha x dx \\ &= \frac{e(\sin \alpha - \alpha \cos \alpha) - e^{-1}(-\sin \alpha - \alpha \cos \alpha)}{1 + \alpha^2} \\ &= \frac{2(\cosh 1) \sin \alpha - 2\alpha(\sinh 1) \cos \alpha}{1 + \alpha^2}. \end{aligned}$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty [A(\alpha) \cos \alpha x + B(\alpha) \sin \alpha x] d\alpha.$$

7. The function is odd. Thus from formula (11) in the text

$$B(\alpha) = 5 \int_0^1 \sin \alpha x dx = \frac{5(1 - \cos \alpha)}{\alpha}.$$

Hence from formula (10) in the text,

$$f(x) = \frac{10}{\pi} \int_0^\infty \frac{(1 - \cos \alpha) \sin \alpha x}{\alpha} d\alpha.$$

8. The function is even. Thus from formula (9) in the text

$$A(\alpha) = \pi \int_1^2 \cos \alpha x dx = \pi \left(\frac{\sin 2\alpha - \sin \alpha}{\alpha} \right).$$

Hence from formula (8) in the text,

$$f(x) = 2 \int_0^\infty \frac{(\sin 2\alpha - \sin \alpha) \cos \alpha x}{\alpha} d\alpha.$$

9. The function is even. Thus from formula (9) in the text

$$\begin{aligned} A(\alpha) &= \int_0^\pi x \cos \alpha x dx = \frac{x \sin \alpha x}{\alpha} \Big|_0^\pi - \frac{1}{\alpha} \int_0^\pi \sin \alpha x dx \\ &= \frac{\pi \alpha \sin \pi \alpha}{\alpha} + \frac{1}{\alpha^2} \cos \alpha x \Big|_0^\pi = \frac{\pi \alpha \sin \pi \alpha + \cos \pi \alpha - 1}{\alpha^2}. \end{aligned}$$

15.3 Fourier Integral

Hence from formula (8) in the text

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(\pi\alpha \sin \pi\alpha + \cos \pi\alpha - 1) \cos \alpha x}{\alpha^2} d\alpha.$$

10. The function is odd. Thus from formula (11) in the text

$$\begin{aligned} B(\alpha) &= \int_0^\pi x \sin \alpha x dx = -\frac{x \cos \alpha x}{\alpha} \Big|_0^\pi + \frac{1}{\alpha} \int_0^\pi \cos \alpha x dx \\ &= -\frac{\pi \cos \pi\alpha}{\alpha} + \frac{1}{\alpha^2} \sin \alpha x \Big|_0^\pi = \frac{-\pi\alpha \cos \pi\alpha + \sin \pi\alpha}{\alpha^2}. \end{aligned}$$

Hence from formula (10) in the text,

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(-\pi\alpha \cos \pi\alpha + \sin \pi\alpha) \sin \alpha x}{\alpha^2} d\alpha.$$

11. The function is odd. Thus from formula (11) in the text

$$\begin{aligned} B(\alpha) &= \int_0^\infty (e^{-x} \sin x) \sin \alpha x dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} [\cos(1-\alpha)x - \cos(1+\alpha)x] dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} \cos(1-\alpha)x dx - \frac{1}{2} \int_0^\infty e^{-x} \cos(1+\alpha)x dx. \end{aligned}$$

Now recall

$$\mathcal{L}\{\cos kt\} = \int_0^\infty e^{-st} \cos kt dt = s/(s^2 + k^2).$$

If we set $s = 1$, and in turn, $k = 1 - \alpha$ and then $k = 1 + \alpha$, we obtain

$$B(\alpha) = \frac{1}{2} \frac{1}{1 + (1 - \alpha)^2} - \frac{1}{2} \frac{1}{1 + (1 + \alpha)^2} = \frac{1}{2} \frac{(1 + \alpha)^2 - (1 - \alpha)^2}{[1 + (1 - \alpha)^2][1 + (1 + \alpha)^2]}.$$

Simplifying the last expression gives

$$B(\alpha) = \frac{2\alpha}{4 + \alpha^4}.$$

Hence from formula (10) in the text

$$f(x) = \frac{4}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{4 + \alpha^4} d\alpha.$$

12. The function is odd. Thus from formula (11) in the text

$$B(\alpha) = \int_0^\infty x e^{-x} \sin \alpha x dx.$$

Now recall

$$\mathcal{L}\{t \sin kt\} = -\frac{d}{ds} \mathcal{L}\{\sin kt\} = 2ks/(s^2 + k^2)^2.$$

If we set $s = 1$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{2\alpha}{(1 + \alpha^2)^2}.$$

Hence from formula (10) in the text

$$f(x) = \frac{4}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(1 + \alpha^2)^2} d\alpha.$$

13. For the cosine integral,

$$A(\alpha) = \int_0^\infty e^{-kx} \cos \alpha x \, dx = \frac{k}{k^2 + \alpha^2}.$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{k \cos \alpha x}{k^2 + \alpha^2} d\alpha = \frac{2k}{\pi} \int_0^\infty \frac{\cos \alpha x}{k^2 + \alpha^2} d\alpha.$$

For the sine integral,

$$B(\alpha) = \int_0^\infty e^{-kx} \sin \alpha x \, dx = \frac{\alpha}{k^2 + \alpha^2}.$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{k^2 + \alpha^2} d\alpha.$$

14. From Problem 13 the cosine and sine integral representations of e^{-kx} , $k > 0$, are respectively,

$$e^{-kx} = \frac{2k}{\pi} \int_0^\infty \frac{\cos \alpha x}{k^2 + \alpha^2} d\alpha \quad \text{and} \quad e^{-kx} = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{k^2 + \alpha^2} d\alpha.$$

Hence, the cosine integral representation of $f(x) = e^{-x} - e^{-3x}$ is

$$e^{-x} - e^{-3x} = \frac{2}{\pi} \int_0^\infty \frac{\cos \alpha x}{1 + \alpha^2} d\alpha - \frac{2(3)}{\pi} \int_0^\infty \frac{\cos \alpha x}{9 + \alpha^2} d\alpha = \frac{4}{\pi} \int_0^\infty \frac{3 - \alpha^2}{(1 + \alpha^2)(9 + \alpha^2)} \cos \alpha x \, d\alpha.$$

The sine integral representation of f is

$$e^{-x} - e^{-3x} = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{1 + \alpha^2} d\alpha - \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{9 + \alpha^2} d\alpha = \frac{16}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(1 + \alpha^2)(9 + \alpha^2)} d\alpha.$$

15. For the cosine integral,

$$A(\alpha) = \int_0^\infty x e^{-2x} \cos \alpha x \, dx.$$

But we know

$$\mathcal{L}\{t \cos kt\} = -\frac{d}{ds} \frac{s}{(s^2 + k^2)} = \frac{(s^2 - k^2)}{(s^2 + k^2)^2}.$$

If we set $s = 2$ and $k = \alpha$ we obtain

$$A(\alpha) = \frac{4 - \alpha^2}{(4 + \alpha^2)^2}.$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(4 - \alpha^2) \cos \alpha x}{(4 + \alpha^2)^2} d\alpha.$$

For the sine integral,

$$B(\alpha) = \int_0^\infty x e^{-2x} \sin \alpha x \, dx.$$

From Problem 12, we know

$$\mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}.$$

If we set $s = 2$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{4\alpha}{(4 + \alpha^2)^2}.$$

Hence

$$f(x) = \frac{8}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(4 + \alpha^2)^2} d\alpha.$$

15.3 Fourier Integral

16. For the cosine integral,

$$\begin{aligned}
 A(\alpha) &= \int_0^\infty e^{-x} \cos x \cos \alpha x \, dx \\
 &= \frac{1}{2} \int_0^\infty e^{-x} [\cos(1+\alpha)x + \cos(1-\alpha)x] \, dx \\
 &= \frac{1}{2} \frac{1}{1+(1+\alpha)^2} + \frac{1}{2} \frac{1}{1+(1-\alpha)^2} \\
 &= \frac{1}{2} \frac{1+(1-\alpha)^2 + 1+(1+\alpha)^2}{[1+(1+\alpha)^2][1+(1-\alpha)^2]} \\
 &= \frac{2+\alpha^2}{4+\alpha^4}.
 \end{aligned}$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(2+\alpha^2) \cos \alpha x}{4+\alpha^4} \, d\alpha.$$

For the sine integral,

$$\begin{aligned}
 B(\alpha) &= \int_0^\infty e^{-x} \cos x \sin \alpha x \, dx \\
 &= \frac{1}{2} \int_0^\infty e^{-x} [\sin(1+\alpha)x - \sin(1-\alpha)x] \, dx \\
 &= \frac{1}{2} \frac{1+\alpha}{1+(1+\alpha)^2} - \frac{1}{2} \frac{1-\alpha}{1+(1-\alpha)^2} \\
 &= \frac{1}{2} \left[\frac{(1+\alpha)[1+(1-\alpha)^2] - (1-\alpha)[1+(1+\alpha)^2]}{[1+(1+\alpha)^2][1+(1-\alpha)^2]} \right] \\
 &= \frac{\alpha^3}{4+\alpha^4}.
 \end{aligned}$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{\alpha^3 \sin \alpha x}{4+\alpha^4} \, d\alpha.$$

17. By formula (8) in the text

$$f(x) = 2\pi \int_0^\infty e^{-\alpha} \cos \alpha x \, d\alpha = \frac{2}{\pi} \frac{1}{1+x^2}, \quad x > 0.$$

18. From the formula for sine integral of $f(x)$ we have

$$\begin{aligned}
 f(x) &= \frac{2}{\pi} \int_0^\infty \left(\int_0^\infty f(x) \sin \alpha x \, dx \right) \sin \alpha x \, dx \\
 &= \frac{2}{\pi} \left[\int_0^1 1 \cdot \sin \alpha x \, d\alpha + \int_1^\infty 0 \cdot \sin \alpha x \, d\alpha \right] \\
 &= \frac{2}{\pi} \frac{(-\cos \alpha x)}{x} \Big|_0^1 = \frac{2}{\pi} \frac{1 - \cos x}{x}.
 \end{aligned}$$

19. (a) From formula (7) in the text with $x = 2$, we have

$$\frac{1}{2} = \frac{2}{\pi} \int_0^\infty \frac{\sin \alpha \cos \alpha}{\alpha} \, d\alpha = \frac{1}{\pi} \int_0^\infty \frac{\sin 2\alpha}{\alpha} \, d\alpha.$$

If we let $\alpha = x$ we obtain

$$\int_0^\infty \frac{\sin 2x}{x} dx = \frac{\pi}{2}.$$

(b) If we now let $2x = kt$ where $k > 0$, then $dx = (k/2)dt$ and the integral in part (a) becomes

$$\int_0^\infty \frac{\sin kt}{kt/2} (k/2) dt = \int_0^\infty \frac{\sin kt}{t} dt = \frac{\pi}{2}.$$

20. With $f(x) = e^{-|x|}$, formula (16) in the text is

$$C(\alpha) = \int_{-\infty}^\infty e^{-|x|} e^{i\alpha x} dx = \int_{-\infty}^\infty e^{-|x|} \cos \alpha x dx + i \int_{-\infty}^\infty e^{-|x|} \sin \alpha x dx.$$

The imaginary part in the last line is zero since the integrand is an odd function of x . Therefore,

$$C(\alpha) = \int_{-\infty}^\infty e^{-|x|} \cos \alpha x dx = 2 \int_0^\infty e^{-x} \cos \alpha x dx = \frac{2}{1 + \alpha^2}$$

and so from formula (15) in the text,

$$f(x) = \frac{1}{\pi} \int_{-\infty}^\infty \frac{\cos \alpha x}{1 + \alpha^2} d\alpha = \frac{2}{\pi} \int_0^\infty \frac{\cos \alpha x}{1 + \alpha^2} d\alpha.$$

This is the same result obtained from formulas (8) and (9) in the text.

21. (a) From the identity

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

we have

$$\begin{aligned} \sin \alpha \cos \alpha x &= \frac{1}{2} [\sin(\alpha + \alpha x) + \sin(\alpha - \alpha x)] \\ &= \frac{1}{2} [\sin \alpha(1 + x) + \sin \alpha(1 - x)] \\ &= \frac{1}{2} [\sin \alpha(x + 1) - \sin \alpha(x - 1)]. \end{aligned}$$

Then

$$\frac{2}{\pi} \int_0^\infty \frac{\sin \alpha \cos \alpha x}{\alpha} d\alpha = \frac{1}{\pi} \int_0^\infty \frac{\sin \alpha(x + 1) - \sin \alpha(x - 1)}{\alpha} d\alpha.$$

(b) Noting that

$$\begin{aligned} F_b &= \frac{1}{\pi} \int_0^b \frac{\sin \alpha(x + 1) - \sin \alpha(x - 1)}{\alpha} d\alpha \\ &= \frac{1}{\pi} \left[\int_0^b \frac{\sin \alpha(x + 1)}{\alpha} d\alpha - \int_0^b \frac{\sin \alpha(x - 1)}{\alpha} d\alpha \right] \end{aligned}$$

and letting $t = \alpha(x + 1)$ so that $dt = (x + 1) d\alpha$ in the first integral and $t = \alpha(x - 1)$ so that $dt = (x - 1) d\alpha$ in the second integral we have

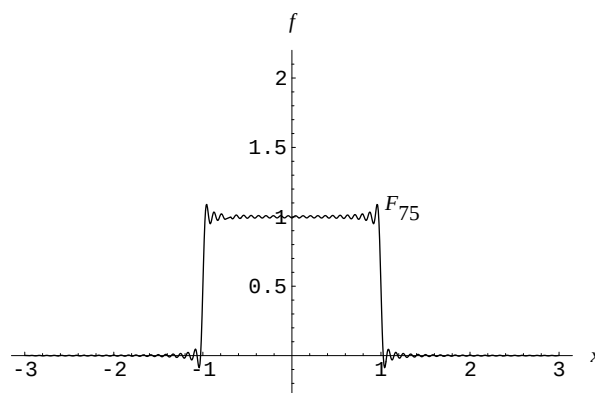
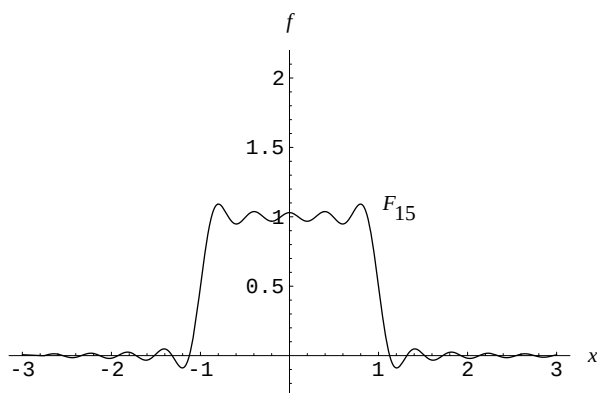
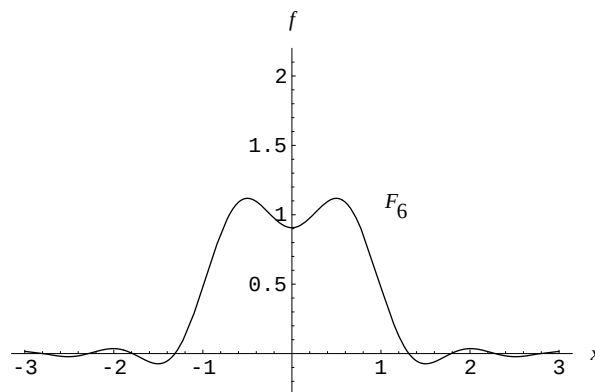
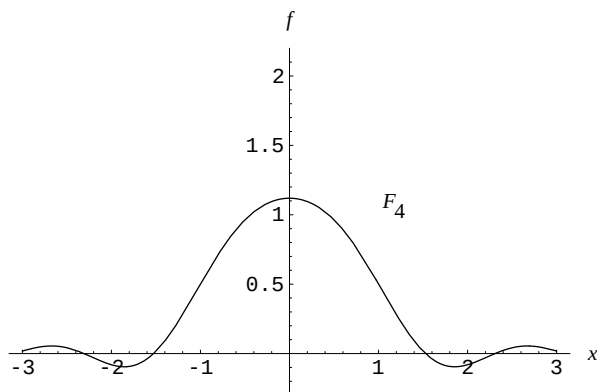
$$F_b = \frac{1}{\pi} \left[\int_0^{b(x+1)} \frac{\sin t}{t} dt - \int_0^{b(x-1)} \frac{\sin t}{t} dt \right].$$

Since $\text{Si}(x) = \int_0^x [(\sin t)/t] dt$, this becomes

$$F_b = \frac{1}{\pi} [\text{Si}(b(x + 1)) - \text{Si}(b(x - 1))].$$

15.3 Fourier Integral

(c) In *Mathematica* we define $f[b_] := (1/\text{Pi})(\text{SinIntegral}[b(x+1)] - \text{SinIntegral}[b(x-1)])$. Graphs of $F_b(x)$ for $b = 4, 6, 15$, and 75 are shown below.



EXERCISES 15.4

Fourier Transforms

For the boundary-value problems in this section it is sometimes useful to note that the identities

$$e^{i\alpha} = \cos \alpha + i \sin \alpha \quad \text{and} \quad e^{-i\alpha} = \cos \alpha - i \sin \alpha$$

imply

$$e^{i\alpha} + e^{-i\alpha} = 2 \cos \alpha \quad \text{and} \quad e^{i\alpha} - e^{-i\alpha} = 2i \sin \alpha.$$

- Using the Fourier transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = 0 \quad \text{and so} \quad U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = \mathcal{F}\{e^{-|x|}\}.$$

We have

$$\mathcal{F}\{e^{-|x|}\} = \int_{-\infty}^{\infty} e^{-|x|} e^{i\alpha x} dx = \int_{-\infty}^{\infty} e^{-|x|} (\cos \alpha x + i \sin \alpha x) dx = \int_{-\infty}^{\infty} e^{-|x|} \cos \alpha x dx.$$

The integral

$$\int_{-\infty}^{\infty} e^{-|x|} \sin \alpha x dx = 0$$

since the integrand is an odd function of x . Continuing we obtain

$$\mathcal{F}\{e^{-|x|}\} = 2 \int_0^{\infty} e^{-x} \cos \alpha x dx = \frac{2}{1 + \alpha^2}.$$

But $U(\alpha, 0) = c = 2/(1 + \alpha^2)$ gives

$$U(\alpha, t) = \frac{2e^{-k\alpha^2 t}}{1 + \alpha^2}$$

and so

$$\begin{aligned} u(x, t) &= \frac{2}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t} e^{-i\alpha x}}{1 + \alpha^2} d\alpha = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t}}{1 + \alpha^2} (\cos \alpha x - i \sin \alpha x) d\alpha \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t} \cos \alpha x}{1 + \alpha^2} d\alpha = \frac{2}{\pi} \int_0^{\infty} \frac{e^{-k\alpha^2 t} \cos \alpha x}{1 + \alpha^2} d\alpha. \end{aligned}$$

2. Using the Fourier sine transform we find $U(\alpha, t) = ce^{-k\alpha^2 t}$. The Fourier sine transform of the initial condition is

$$\mathcal{F}_S\{u(x, 0)\} = \int_0^{\infty} u(x, 0) \sin \alpha x dx = \int_0^1 100 \sin \alpha x dx = \frac{100}{\alpha} (1 - \cos \alpha).$$

Thus $U(\alpha, 0) = (100/\alpha)(1 - \cos \alpha)$ and since $c = U(\alpha, 0)$, we have

$$U(\alpha, t) = \frac{100}{\alpha} (1 - \cos \alpha) e^{-k\alpha^2 t}.$$

Applying the inverse Fourier transform we obtain

$$\begin{aligned} u(x, t) &= \mathcal{F}_S^{-1}\{U(\alpha, t)\} = \frac{2}{\pi} \int_0^{\infty} \frac{100}{\alpha} (1 - \cos \alpha) e^{-k\alpha^2 t} \sin \alpha x d\alpha \\ &= \frac{200}{\pi} \int_0^{\infty} \frac{1 - \cos \alpha}{\alpha} e^{-k\alpha^2 t} \sin \alpha x dx. \end{aligned}$$

3. Using the Fourier sine transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = k\alpha u_0.$$

The general solution of this linear equation is

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{u_0}{\alpha}.$$

But $U(\alpha, 0) = 0$ implies $c = -u_0/\alpha$ and so

$$U(\alpha, t) = u_0 \frac{1 - e^{-k\alpha^2 t}}{\alpha}$$

and

$$u(x, t) = \frac{2u_0}{\pi} \int_0^{\infty} \frac{1 - e^{-k\alpha^2 t}}{\alpha} \sin \alpha x d\alpha.$$

4. The solution of Problem 3 can be written

$$u(x, t) = \frac{2u_0}{\pi} \int_0^{\infty} \frac{\sin \alpha x}{\alpha} d\alpha - \frac{2u_0}{\pi} \int_0^{\infty} \frac{\sin \alpha x}{\alpha} e^{-k\alpha^2 t} d\alpha.$$

15.4 Fourier Transforms

Using

$$\int_0^\infty \frac{\sin \alpha x}{\alpha} d\alpha = \pi/2$$

the last line becomes

$$u(x, t) = u_0 - \frac{2u_0}{\pi} \int_0^\infty \frac{\sin \alpha x}{\alpha} e^{-k\alpha^2 t} d\alpha.$$

5. Using the Fourier sine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = U(\alpha, 0) = \int_0^1 \sin \alpha x dx = \frac{1 - \cos \alpha}{\alpha}.$$

From this we find $c = (1 - \cos \alpha)/\alpha$ and so

$$U(\alpha, t) = \frac{1 - \cos \alpha}{\alpha} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos \alpha}{\alpha} e^{-k\alpha^2 t} \sin \alpha x d\alpha.$$

6. Since the domain of x is $(0, \infty)$ and the condition at $x = 0$ involves $\partial u / \partial x$ we use the Fourier cosine transform:

$$-k\alpha^2 U(\alpha, t) - ku_x(0, t) = \frac{dU}{dt}$$

$$\frac{dU}{dt} + k\alpha^2 U = kA$$

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{A}{\alpha^2}.$$

Since

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = 0$$

we find $c = -A/\alpha^2$, so that

$$U(\alpha, t) = A \frac{1 - e^{-k\alpha^2 t}}{\alpha^2}.$$

Applying the inverse Fourier cosine transform we obtain

$$u(x, t) = \mathcal{F}_C^{-1}\{U(\alpha, t)\} = \frac{2A}{\pi} \int_0^\infty \frac{1 - e^{-k\alpha^2 t}}{\alpha^2} \cos \alpha x d\alpha.$$

7. Using the Fourier cosine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}_C\{u(x, 0)\} = \int_0^1 \cos \alpha x dx = \frac{\sin \alpha}{\alpha} = U(\alpha, 0).$$

From this we obtain $c = (\sin \alpha)/\alpha$ and so

$$U(\alpha, t) = \frac{\sin \alpha}{\alpha} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \frac{\sin \alpha}{\alpha} e^{-k\alpha^2 t} \cos \alpha x d\alpha.$$

8. Using the Fourier sine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{1}{\alpha}.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = \mathcal{F}_S\{e^{-x}\} = \int_0^\infty e^{-x} \sin \alpha x \, dx = \frac{\alpha}{1 + \alpha^2} = U(\alpha, 0).$$

From this we obtain $c = \alpha/(1 + \alpha^2) - 1/\alpha$. Therefore

$$U(\alpha, t) = \left(\frac{\alpha}{1 + \alpha^2} - \frac{1}{\alpha} \right) e^{-k\alpha^2 t} + \frac{1}{\alpha} = \frac{1}{\alpha} - \frac{e^{-k\alpha^2 t}}{\alpha(1 + \alpha^2)}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \left(\frac{1}{\alpha} - \frac{e^{-k\alpha^2 t}}{\alpha(1 + \alpha^2)} \right) \sin \alpha x \, d\alpha.$$

9. (a) Using the Fourier transform we obtain

$$U(\alpha, t) = c_1 \cos \alpha at + c_2 \sin \alpha at.$$

If we write

$$\mathcal{F}\{u(x, 0)\} = \mathcal{F}\{f(x)\} = F(\alpha)$$

and

$$\mathcal{F}\{u_t(x, 0)\} = \mathcal{F}\{g(x)\} = G(\alpha)$$

we first obtain $c_1 = F(\alpha)$ from $U(\alpha, 0) = F(\alpha)$ and then $c_2 = G(\alpha)/\alpha a$ from $dU/dt|_{t=0} = G(\alpha)$. Thus

$$U(\alpha, t) = F(\alpha) \cos \alpha at + \frac{G(\alpha)}{\alpha a} \sin \alpha at$$

and

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^\infty \left(F(\alpha) \cos \alpha at + \frac{G(\alpha)}{\alpha a} \sin \alpha at \right) e^{-i\alpha x} d\alpha.$$

(b) If $g(x) = 0$ then $c_2 = 0$ and

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^\infty F(\alpha) \cos \alpha at e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^\infty F(\alpha) \left(\frac{e^{\alpha at i} + e^{-\alpha at i}}{2} \right) e^{-i\alpha x} d\alpha \\ &= \frac{1}{2} \left[\frac{1}{2\pi} \int_{-\infty}^\infty F(\alpha) e^{-i(x-at)\alpha} d\alpha + \frac{1}{2\pi} \int_{-\infty}^\infty F(\alpha) e^{-i(x+at)\alpha} d\alpha \right] \\ &= \frac{1}{2} [f(x-at) + f(x+at)]. \end{aligned}$$

10. Using the Fourier sine transform we obtain

$$U(\alpha, t) = c_1 \cos \alpha at + c_2 \sin \alpha at.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = \mathcal{F}_S\{xe^{-x}\} = \int_0^\infty xe^{-x} \sin \alpha x \, dx = \frac{2\alpha}{(1 + \alpha^2)^2} = U(\alpha, 0).$$

Also,

$$\mathcal{F}_S\{u_t(x, 0)\} = \frac{dU}{dt} \Big|_{t=0} = 0.$$

15.4 Fourier Transforms

This last condition gives $c_2 = 0$. Then $U(\alpha, 0) = 2\alpha/(1 + \alpha^2)^2$ yields $c_1 = 2\alpha/(1 + \alpha^2)^2$. Therefore

$$U(\alpha, t) = \frac{2\alpha}{(1 + \alpha^2)^2} \cos \alpha at$$

and

$$u(x, t) = \frac{4}{\pi} \int_0^\infty \frac{\alpha \cos \alpha at}{(1 + \alpha^2)^2} \sin \alpha x \, d\alpha.$$

11. Using the Fourier cosine transform we obtain

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

Now the Fourier cosine transforms of $u(0, y) = e^{-y}$ and $u(\pi, y) = 0$ are, respectively, $U(0, \alpha) = 1/(1 + \alpha^2)$ and $U(\pi, \alpha) = 0$. The first of these conditions gives $c_1 = 1/(1 + \alpha^2)$. The second condition gives

$$c_2 = -\frac{\cosh \alpha \pi}{(1 + \alpha^2) \sinh \alpha \pi}.$$

Hence

$$U(x, \alpha) = \frac{\cosh \alpha x}{1 + \alpha^2} - \frac{\cosh \alpha \pi \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} = \frac{\sinh \alpha \pi \cosh \alpha \pi - \cosh \alpha \pi \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} = \frac{\sinh \alpha(\pi - x)}{(1 + \alpha^2) \sinh \alpha \pi}$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\sinh \alpha(\pi - x)}{(1 + \alpha^2) \sinh \alpha \pi} \cos \alpha y \, d\alpha.$$

12. Since the boundary condition at $y = 0$ now involves $u(x, 0)$ rather than $u'(x, 0)$, we use the Fourier sine transform. The transform of the partial differential equation is then

$$\frac{d^2 U}{dx^2} - \alpha^2 U + \alpha u(x, 0) = 0 \quad \text{or} \quad \frac{d^2 U}{dx^2} - \alpha^2 U = -\alpha.$$

The solution of this differential equation is

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x + \frac{1}{\alpha}.$$

The transforms of the boundary conditions at $x = 0$ and $x = \pi$ in turn imply that $c_1 = 1/\alpha$ and

$$c_2 = \frac{\cosh \alpha \pi}{\alpha \sinh \alpha \pi} - \frac{1}{\alpha \sinh \alpha \pi} + \frac{\alpha}{(1 + \alpha^2) \sinh \alpha \pi}.$$

Hence

$$\begin{aligned} U(x, \alpha) &= \frac{1}{\alpha} - \frac{\cosh \alpha x}{\alpha} + \frac{\cosh \alpha \pi}{\alpha \sinh \alpha \pi} \sinh \alpha x - \frac{\sinh \alpha x}{\alpha \sinh \alpha \pi} + \frac{\alpha \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} \\ &= \frac{1}{\alpha} - \frac{\sinh \alpha(\pi - x)}{\alpha \sinh \alpha \pi} - \frac{\sinh \alpha x}{\alpha(1 + \alpha^2) \sinh \alpha \pi}. \end{aligned}$$

Taking the inverse transform it follows that

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \left(\frac{1}{\alpha} - \frac{\sinh \alpha(\pi - x)}{\alpha \sinh \alpha \pi} - \frac{\sinh \alpha x}{\alpha(1 + \alpha^2) \sinh \alpha \pi} \right) \sin \alpha y \, d\alpha.$$

13. Using the Fourier cosine transform with respect to x gives

$$U(\alpha, y) = c_1 e^{-\alpha y} + c_2 e^{\alpha y}.$$

Since we expect $u(x, y)$ to be bounded as $y \rightarrow \infty$ we define $c_2 = 0$. Thus

$$U(\alpha, y) = c_1 e^{-\alpha y}.$$

Now

$$\mathcal{F}_C\{u(x, 0)\} = \int_0^1 50 \cos \alpha x \, dx = 50 \frac{\sin \alpha}{\alpha}$$

and so

$$U(\alpha, y) = 50 \frac{\sin \alpha}{\alpha} e^{-\alpha y}$$

and

$$u(x, y) = \frac{100}{\pi} \int_0^\infty \frac{\sin \alpha}{\alpha} e^{-\alpha y} \cos \alpha x \, d\alpha.$$

14. The boundary condition $u(0, y) = 0$ indicates that we now use the Fourier sine transform. We still have $U(\alpha, y) = c_1 e^{-\alpha y}$, but

$$\mathcal{F}_S\{u(x, 0)\} = \int_0^1 50 \sin \alpha x \, dx = 50(1 - \cos \alpha)/\alpha = U(\alpha, 0).$$

This gives $c_1 = 50(1 - \cos \alpha)/\alpha$ and so

$$U(\alpha, y) = 50 \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y}$$

and

$$u(x, y) = \frac{100}{\pi} \int_0^\infty \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y} \sin \alpha x \, d\alpha.$$

15. We use the Fourier sine transform with respect to x to obtain

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y.$$

The transforms of $u(x, 0) = f(x)$ and $u(x, 2) = 0$ give, in turn, $U(\alpha, 0) = F(\alpha)$ and $U(\alpha, 2) = 0$. The first condition gives $c_1 = F(\alpha)$ and the second condition then yields

$$c_2 = -\frac{F(\alpha) \cosh 2\alpha}{\sinh 2\alpha}.$$

Hence

$$\begin{aligned} U(\alpha, y) &= F(\alpha) \cosh \alpha y - \frac{F(\alpha) \cosh 2\alpha \sinh \alpha y}{\sinh 2\alpha} \\ &= F(\alpha) \frac{\sinh 2\alpha \cosh \alpha y - \cosh 2\alpha \sinh \alpha y}{\sinh 2\alpha} \\ &= F(\alpha) \frac{\sinh \alpha(2 - y)}{\sinh 2\alpha} \end{aligned}$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty F(\alpha) \frac{\sinh \alpha(2 - y)}{\sinh 2\alpha} \sin \alpha x \, d\alpha.$$

16. The domain of y and the boundary condition at $y = 0$ suggest that we use a Fourier cosine transform. The transformed equation is

$$\frac{d^2 U}{dx^2} - \alpha^2 U - u_y(x, 0) = 0 \quad \text{or} \quad \frac{d^2 U}{dx^2} - \alpha^2 U = 0.$$

Because the domain of the variable x is a finite interval we choose to write the general solution of the latter equation as

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

15.4 Fourier Transforms

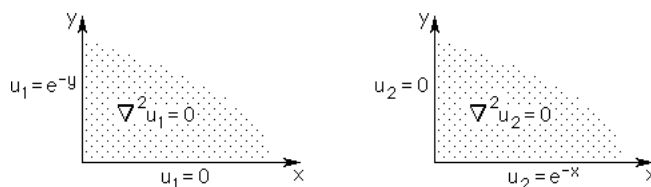
Now $U(0, \alpha) = F(\alpha)$, where $F(\alpha)$ is the Fourier cosine transform of $f(y)$, and $U'(\pi, \alpha) = 0$ imply $c_1 = F(\alpha)$ and $c_2 = -F(\alpha) \sinh \alpha\pi / \cosh \alpha\pi$. Thus

$$U(x, \alpha) = F(\alpha) \cosh \alpha x - F(\alpha) \frac{\sinh \alpha\pi}{\cosh \alpha\pi} \sinh \alpha x = F(\alpha) \frac{\cosh \alpha(\pi - x)}{\cosh \alpha\pi}.$$

Using the inverse transform we find that a solution to the problem is

$$u(x, y) = \frac{2}{\pi} \int_0^\infty F(\alpha) \frac{\cosh \alpha(\pi - x)}{\cosh \alpha\pi} \cos \alpha y d\alpha.$$

17. We solve two boundary-value problems:



Using the Fourier sine transform with respect to y gives

$$u_1(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha e^{-\alpha x}}{1 + \alpha^2} \sin \alpha y d\alpha.$$

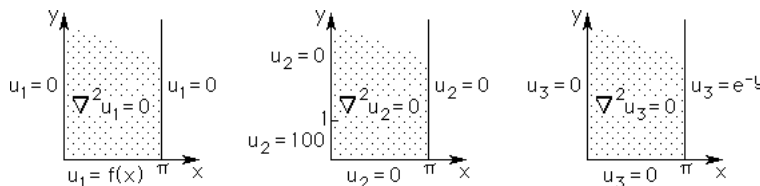
The Fourier sine transform with respect to x yields the solution to the second problem:

$$u_2(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha e^{-\alpha y}}{1 + \alpha^2} \sin \alpha x d\alpha.$$

We define the solution of the original problem to be

$$u(x, y) = u_1(x, y) + u_2(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha}{1 + \alpha^2} [e^{-\alpha x} \sin \alpha y + e^{-\alpha y} \sin \alpha x] d\alpha.$$

18. We solve the three boundary-value problems:



Using separation of variables we find the solution of the first problem is

$$u_1(x, y) = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx \quad \text{where} \quad A_n = \frac{2}{\pi} \int_0^\pi f(x) \sin nx dx.$$

Using the Fourier sine transform with respect to y gives the solution of the second problem:

$$u_2(x, y) = \frac{200}{\pi} \int_0^\infty \frac{(1 - \cos \alpha) \sinh \alpha(\pi - x)}{\alpha \sinh \alpha\pi} \sin \alpha y d\alpha.$$

Also, the Fourier sine transform with respect to y gives the solution of the third problem:

$$u_3(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha\pi} \sin \alpha y d\alpha.$$

The solution of the original problem is

$$u(x, y) = u_1(x, y) + u_2(x, y) + u_3(x, y).$$

19. Using the Fourier transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = 0 \quad \text{and so} \quad U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = \sqrt{\pi} e^{-\alpha^2/4}$$

by the given result. This gives $c = \sqrt{\pi} e^{-\alpha^2/4}$ and so

$$U(\alpha, t) = \sqrt{\pi} e^{-(\frac{1}{4} + kt)\alpha^2}.$$

Using the given Fourier transform again we obtain

$$u(x, t) = \sqrt{\pi} \mathcal{F}^{-1}\{e^{-(1+4kt)\alpha^2/4}\} = \frac{1}{\sqrt{1+4kt}} e^{-x^2/(1+4kt)}.$$

20. We use $U(\alpha, t) = ce^{-k\alpha^2 t}$. The Fourier transform of the boundary condition is $U(\alpha, 0) = F(\alpha)$. This gives $c = F(\alpha)$ and so $U(\alpha, t) = F(\alpha)e^{-k\alpha^2 t}$. By the convolution theorem and the given result, we obtain

$$u(x, t) = \mathcal{F}^{-1}\{F(\alpha) \cdot e^{-k\alpha^2 t}\} = \frac{1}{2\sqrt{k\pi t}} \int_{-\infty}^{\infty} f(\tau) e^{-(x-\tau)^2/4kt} d\tau.$$

21. Using the Fourier transform with respect to x gives

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y.$$

The transform of the boundary condition $\partial u / \partial y \big|_{y=0} = 0$ is $dU/dy \big|_{y=0} = 0$. This condition gives $c_2 = 0$. Hence

$$U(\alpha, y) = c_1 \cosh \alpha y.$$

Now by the given information the transform of the boundary condition $u(x, 1) = e^{-x^2}$ is $U(\alpha, 1) = \sqrt{\pi} e^{-\alpha^2/4}$. This condition then gives $c_1 = \sqrt{\pi} e^{-\alpha^2/4} \cosh \alpha$. Therefore

$$U(\alpha, y) = \sqrt{\pi} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha}$$

and

$$\begin{aligned} U(x, y) &= \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} e^{-i\alpha x} d\alpha = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} \cos \alpha x d\alpha \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} \cos \alpha x d\alpha. \end{aligned}$$

22. From the Table of Laplace transforms we have

$$\int_0^{\infty} e^{-st} \frac{\sin at}{t} dt = \arctan \frac{a}{s}$$

and

$$\int_0^{\infty} e^{-st} \frac{\sin at \cos bt}{t} dt = \frac{1}{2} \arctan \frac{a+b}{s} + \frac{1}{2} \arctan \frac{a-b}{s}.$$

Identifying $\alpha = t$, $x = a$, and $y = s$, the solution of Problem 14 is

$$\begin{aligned} u(x, y) &= \frac{100}{\pi} \int_0^{\infty} \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y} \sin \alpha x d\alpha \\ &= \frac{100}{\pi} \left[\int_0^{\infty} \frac{\sin \alpha x}{\alpha} e^{-\alpha y} d\alpha - \int_0^{\infty} \frac{\sin \alpha x \cos \alpha}{\alpha} e^{-\alpha y} d\alpha \right] \\ &= \frac{100}{\pi} \left[\arctan \frac{x}{y} - \frac{1}{2} \arctan \frac{x+1}{y} - \frac{1}{2} \arctan \frac{x-1}{y} \right]. \end{aligned}$$

15.4 Fourier Transforms

23. Using the definition of f and the solution in Problem 20 we obtain

$$u(x, t) = \frac{u_0}{2\sqrt{k\pi t}} \int_{-1}^1 e^{-(x-\tau)^2/4kt} d\tau.$$

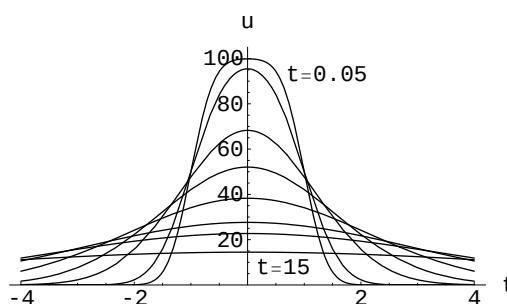
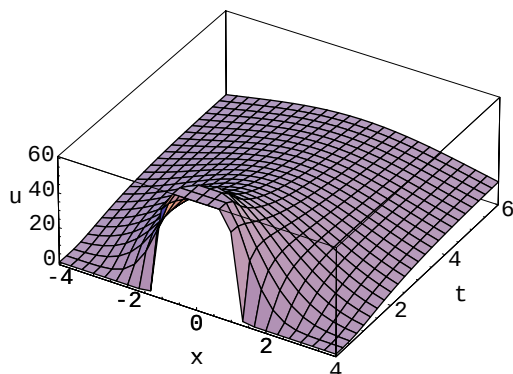
If $v = (x - \tau)/2\sqrt{kt}$, then $d\tau = -2\sqrt{kt} du$ and the integral becomes

$$v(x, t) = \frac{u_0}{\sqrt{\pi}} \int_{(x-1)/2\sqrt{kt}}^{(x+1)/2\sqrt{kt}} e^{-v^2} dv.$$

Using the result in Problem 9 of Exercises 15.1 in the text, we have

$$u(x, t) = \frac{u_0}{2} \left[\operatorname{erf} \left(\frac{x+1}{2\sqrt{kt}} \right) - \operatorname{erf} \left(\frac{x-1}{2\sqrt{kt}} \right) \right].$$

24.



Since $\operatorname{erf}(0) = 0$ and $\lim_{x \rightarrow \infty} \operatorname{erf}(x) = 1$, we have

$$\lim_{t \rightarrow \infty} u(x, t) = 50[\operatorname{erf}(0) - \operatorname{erf}(0)] = 0$$

and

$$\lim_{x \rightarrow \infty} u(x, t) = 50[\operatorname{erf}(\infty) - \operatorname{erf}(\infty)] = 50[1 - 1] = 0.$$

EXERCISES 15.5

Fast Fourier Transform

1. We show that $\frac{1}{4}\overline{F}_4 F_4 = I$:

$$\frac{1}{4}\overline{F}_4 F_4 = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} = I.$$

Thus $F_4^{-1} = \frac{1}{4}\overline{F}_4$.

2. We have

$$\int_{-\infty}^{\infty} f(x) \delta_{\epsilon}(x-a) dx = \frac{1}{2\epsilon} \int_{a-\epsilon}^{a+\epsilon} f(x) dx = \frac{1}{2\epsilon} f(c)(2\epsilon) = f(c)$$

by the mean value theorem for integrals.

3. By the sifting property,

$$\mathcal{F}\{\delta(x)\} = \int_{-\infty}^{\infty} \delta(x)e^{i\alpha x} dx = e^{i\alpha 0} = 1.$$

4. We already know that $f * \delta = \delta * f$. Then, by the sifting property,

$$(f * \delta)(x) = \int_{-\infty}^{\infty} f(\tau)\delta(x - \tau)d\tau = \int_{-\infty}^{\infty} f(\tau)\delta(\tau - x)d\tau = f(x).$$

5. Using integration by parts with $u = f(x)$ and $dv = \delta'(x - a)$ we find

$$\int_{-\infty}^{\infty} f(x)\delta'(x - a)dx = - \int_{-\infty}^{\infty} f'(x)\delta(x - a)dx = -f'(a)$$

by the sifting property.

6. Using a CAS we find

$$\mathcal{F}\{g(x)\} = \frac{1}{2}[\text{sign}(A - \alpha) + \text{sign}(A + \alpha)]$$

where $\text{sign}(t) = 1$ if $t > 0$ and $\text{sign } t = -1$ if $t < 0$. Thus

$$\mathcal{F}\{g(x)\} = \begin{cases} 1, & -A < \alpha < A \\ 0, & \text{elsewhere.} \end{cases}$$

7. Using

$$\begin{aligned} \omega_8 &= \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & \omega_8^5 &= -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ \omega_8^2 &= i & \omega_8^6 &= -i \\ \omega_8^3 &= -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & \omega_8^7 &= \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ \omega_8^4 &= -1 & \omega_8^8 &= 1 \end{aligned}$$

we have

$$F_8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & i & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -1 & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -i & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ 1 & i & -1 & -i & 1 & i & -1 & -i \\ 1 & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -i & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -1 & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & i & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & i & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -1 & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -i & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \\ 1 & -i & -1 & i & 1 & -i & -1 & i \\ 1 & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -i & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -1 & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & i & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \end{pmatrix}.$$

In factored form

$$F_8 = \begin{pmatrix} I_4 & D_4 \\ I_4 & -D_4 \end{pmatrix} \begin{pmatrix} F_4 & 0 \\ 0 & F_4 \end{pmatrix} P,$$

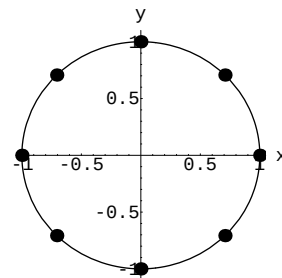
where I_4 is the 4×4 identity matrix.

$$D_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{2}/2 + i\sqrt{2}/2 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -\sqrt{2}/2 + i\sqrt{2}/2 \end{pmatrix},$$

15.5 Fast Fourier Transform

and P is the 8×8 matrix with 1 in positions $(1, 1)$, $(2, 3)$, $(3, 5)$, $(4, 7)$, $(5, 2)$, $(6, 4)$, $(7, 6)$, and $(8, 8)$.

8. The 8th roots of unity, $\omega_8^1, \omega_8^2, \dots, \omega_8^8$ are shown in the solution of Problem 7 above. The points in the complex plane are equally spaced on the perimeter of the unit circle.

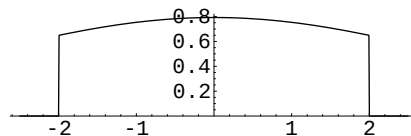


9. The Fourier transform of $g(x) = (\sin 2x)/\pi x$ is

$$G(\alpha) = \begin{cases} 1, & -2 < \alpha < 2 \\ 0, & \text{elsewhere.} \end{cases}$$

This implies that $(f * g)(x) = \mathcal{F}^{-1}\{F(\alpha)G(\alpha)\}$ is band-limited.

The graph of $F(\alpha)G(\alpha)$, which is identical to the graph of $\mathcal{F}(f * g)$, is shown.



10. For $N = 6$,

$$F_6 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1/2 + \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 & -1 & -1/2 - \sqrt{3}i/2 & 1/2 - \sqrt{3}i/2 \\ 1 & -1/2 + \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 & 1 & -1/2 + \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1/2 - \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 & 1 & -1/2 - \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 \\ 1 & 1/2 - \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 & -1 & -1/2 + \sqrt{3}i/2 & 1/2 + \sqrt{3}i/2 \end{pmatrix}.$$

If, for example, $f = (2, 0, 1, 6, 2, 3)$, then

$$c = \frac{1}{6} \overline{F}_6 f = \begin{pmatrix} 7/3 \\ -2/3 + \sqrt{3}i/3 \\ 5/6 - \sqrt{3}i/6 \\ -2/3 \\ 5/6 - \sqrt{3}i/6 \\ -2/3 - i/\sqrt{3} \end{pmatrix}.$$

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1. The partial differential equation and the boundary conditions indicate that the Fourier cosine transform is appropriate for the problem. We find in this case

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\sinh \alpha y}{\alpha(1 + \alpha^2) \cosh \alpha \pi} \cos \alpha x \, d\alpha.$$

2. We use the Laplace transform and undetermined coefficients to obtain

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x + \frac{50}{s + 4\pi^2} \sin 2\pi x.$$

The transformed boundary conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Hence

$$U(x, s) = \frac{50}{s + 4\pi^2} \sin 2\pi x$$

and

$$u(x, t) = 50 \sin 2\pi x \mathcal{L}^{-1} \left\{ \frac{1}{s + 4\pi^2} \right\} = 50e^{-4\pi^2 t} \sin 2\pi x.$$

3. The Laplace transform gives

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + c_2 e^{\sqrt{s+h}x} + \frac{u_0}{s+h}.$$

The condition $\lim_{x \rightarrow \infty} \partial u / \partial x = 0$ implies $\lim_{x \rightarrow \infty} dU/dx = 0$ and so we define $c_2 = 0$. Thus

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + \frac{u_0}{s+h}.$$

The condition $U(0, s) = 0$ then gives $c_1 = -u_0/(s+h)$ and so

$$U(x, s) = \frac{u_0}{s+h} - u_0 \frac{e^{-\sqrt{s+h}x}}{s+h}.$$

With the help of the first translation theorem we then obtain

$$\begin{aligned} u(x, t) &= u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s+h} \right\} - u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s+h}x}}{s+h} \right\} = u_0 e^{-ht} - u_0 e^{-ht} \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) \\ &= u_0 e^{-ht} \left[1 - \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) \right] = u_0 e^{-ht} \operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right). \end{aligned}$$

4. Using the Fourier transform and the result $\mathcal{F}\{e^{-|x|}\} = 1/(1+\alpha^2)$ we find

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} \cos \alpha x d\alpha \\ &= \frac{1}{\pi} \int_0^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} \cos \alpha x d\alpha. \end{aligned}$$

5. The Laplace transform gives

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$ and so we define $c_2 = 0$. Thus

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The transform of the remaining boundary condition is $U(0, s) = 1/s^2$. This gives $c_1 = 1/s^2$. Hence

$$U(x, s) = \frac{e^{-\sqrt{s}x}}{s^2} \quad \text{and} \quad u(x, t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} \frac{e^{-\sqrt{s}x}}{s} \right\}.$$

Using

$$\mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = 1 \quad \text{and} \quad \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} \right\} = \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right),$$

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it follows from the convolution theorem that

$$u(x, t) = \int_0^t \operatorname{erfc}\left(\frac{x}{2\sqrt{\tau}}\right) d\tau.$$

6. The Laplace transform and undetermined coefficients give

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{s-1}{s^2 + \pi^2} \sin \pi x.$$

The conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Thus

$$U(x, s) = \frac{s-1}{s^2 + \pi^2} \sin \pi x$$

and

$$\begin{aligned} u(x, t) &= \sin \pi x \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \pi^2} \right\} - \frac{1}{\pi} \sin \pi x \mathcal{L}^{-1} \left\{ \frac{\pi}{s^2 + \pi^2} \right\} \\ &= (\sin \pi x) \cos \pi t - \frac{1}{\pi} (\sin \pi x) \sin \pi t. \end{aligned}$$

7. The Fourier transform gives the solution

$$\begin{aligned} u(x, t) &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \left(\frac{e^{i\alpha\pi} - 1}{i\alpha} \right) e^{-i\alpha x} e^{-k\alpha^2 t} d\alpha \\ &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{e^{i\alpha(\pi-x)} - e^{-i\alpha x}}{i\alpha} e^{-k\alpha^2 t} d\alpha \\ &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{\cos \alpha(\pi-x) + i \sin \alpha(\pi-x) - \cos \alpha x + i \sin \alpha x}{i\alpha} e^{-k\alpha^2 t} d\alpha. \end{aligned}$$

Since the imaginary part of the integrand of the last integral is an odd function of α , we obtain

$$u(x, t) = \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha(\pi-x) + \sin \alpha x}{\alpha} e^{-k\alpha^2 t} d\alpha.$$

8. Using the Fourier cosine transform we obtain $U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x$. The condition $U(0, \alpha) = 0$ gives $c_1 = 0$. Thus $U(x, \alpha) = c_2 \sinh \alpha x$. Now

$$\mathcal{F}_C\{u(\pi, y)\} = \int_1^2 \cos \alpha y dy = \frac{\sin 2\alpha - \sin \alpha}{\alpha} = U(\pi, \alpha).$$

This last condition gives $c_2 = (\sin 2\alpha - \sin \alpha)/\alpha \sinh \alpha\pi$. Hence

$$U(x, \alpha) = \frac{\sin 2\alpha - \sin \alpha}{\alpha \sinh \alpha\pi} \sinh \alpha x$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\sin 2\alpha - \sin \alpha}{\alpha \sinh \alpha\pi} \sinh \alpha x \cos \alpha y d\alpha.$$

9. We solve the two problems

$$\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0, \quad x > 0, \quad y > 0,$$

$$u_1(0, y) = 0, \quad y > 0,$$

$$u_1(x, 0) = \begin{cases} 100, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$$

and

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0, \quad x > 0, \quad y > 0,$$

$$u_2(0, y) = \begin{cases} 50, & 0 < y < 1 \\ 0, & y > 1 \end{cases}$$

$$u_2(x, 0) = 0.$$

Using the Fourier sine transform with respect to x we find

$$u_1(x, y) = \frac{200}{\pi} \int_0^\infty \left(\frac{1 - \cos \alpha}{\alpha} \right) e^{-\alpha y} \sin \alpha x \, d\alpha.$$

Using the Fourier sine transform with respect to y we find

$$u_2(x, y) = \frac{100}{\pi} \int_0^\infty \left(\frac{1 - \cos \alpha}{\alpha} \right) e^{-\alpha x} \sin \alpha y \, d\alpha.$$

The solution of the problem is then

$$u(x, y) = u_1(x, y) + u_2(x, y).$$

10. The Laplace transform gives

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x + \frac{r}{s^2}.$$

The condition $\partial u / \partial x \big|_{x=0} = 0$ transforms into $dU/dx \big|_{x=0} = 0$. This gives $c_2 = 0$. The remaining condition $u(1, t) = 0$ transforms into $U(1, s) = 0$. This condition then implies $c_1 = -r/s^2 \cosh \sqrt{s}$. Hence

$$U(x, s) = \frac{r}{s^2} - r \frac{\cosh \sqrt{s} x}{s^2 \cosh \sqrt{s}}.$$

Using geometric series and the convolution theorem we obtain

$$\begin{aligned} u(x, t) &= r \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} - r \mathcal{L}^{-1} \left\{ \frac{\cosh \sqrt{s} x}{s^2 \cosh \sqrt{s}} \right\} \\ &= rt - r \sum_{n=0}^{\infty} (-1)^n \left[\int_0^t \operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{\tau}} \right) d\tau + \int_0^t \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{\tau}} \right) d\tau \right]. \end{aligned}$$

11. The Fourier sine transform with respect to x and undetermined coefficients give

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y + \frac{A}{\alpha}.$$

The transforms of the boundary conditions are

$$\frac{dU}{dy} \bigg|_{y=0} = 0 \quad \text{and} \quad \frac{dU}{dy} \bigg|_{y=\pi} = \frac{B\alpha}{1 + \alpha^2}.$$

The first of these conditions gives $c_2 = 0$ and so

$$U(\alpha, y) = c_1 \cosh \alpha y + \frac{A}{\alpha}.$$

The second transformed boundary condition yields $c_1 = B/(1 + \alpha^2) \sinh \alpha \pi$. Therefore

$$U(\alpha, y) = \frac{B \cosh \alpha y}{(1 + \alpha^2) \sinh \alpha \pi} + \frac{A}{\alpha}$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \left(\frac{B \cosh \alpha y}{(1 + \alpha^2) \sinh \alpha \pi} + \frac{A}{\alpha} \right) \sin \alpha x \, d\alpha.$$

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12. Using the Laplace transform gives

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x.$$

The condition $u(0, t) = u_0$ transforms into $U(0, s) = u_0/s$. This gives $c_1 = u_0/s$. The condition $u(1, t) = u_0$ transforms into $U(1, s) = u_0/s$. This implies that $c_2 = u_0(1 - \cosh \sqrt{s})/s \sinh \sqrt{s}$. Hence

$$\begin{aligned} U(x, s) &= \frac{u_0}{s} \cosh \sqrt{s} x + u_0 \left[\frac{1 - \cosh \sqrt{s}}{s \sinh \sqrt{s}} \right] \sinh \sqrt{s} x \\ &= u_0 \left[\frac{\sinh \sqrt{s} \cosh \sqrt{s} x - \cosh \sqrt{s} \sinh \sqrt{s} x + \sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \\ &= u_0 \left[\frac{\sinh \sqrt{s} (1 - x) + \sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \\ &= u_0 \left[\frac{\sinh \sqrt{s} (1 - x)}{s \sinh \sqrt{s}} + \frac{\sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \end{aligned}$$

and

$$\begin{aligned} u(x, t) &= u_0 \left[\mathcal{L}^{-1} \left\{ \frac{\sinh \sqrt{s} (1 - x)}{s \sinh \sqrt{s}} \right\} + \mathcal{L}^{-1} \left\{ \frac{\sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right\} \right] \\ &= u_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+2-x}{2\sqrt{t}} \right) - \operatorname{erf} \left(\frac{2n+x}{2\sqrt{t}} \right) \right] \\ &\quad + u_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+1+x}{2\sqrt{t}} \right) - \operatorname{erf} \left(\frac{2n+1-x}{2\sqrt{t}} \right) \right]. \end{aligned}$$

13. Using the Fourier transform gives

$$U(\alpha, t) = c_1 e^{-k\alpha^2 t}.$$

Now

$$u(\alpha, 0) = \int_0^{\infty} e^{-x} e^{i\alpha x} dx = \frac{e^{(i\alpha-1)x}}{i\alpha-1} \Big|_0^{\infty} = 0 - \frac{1}{i\alpha-1} = \frac{1}{1-i\alpha} = c_1$$

so

$$U(\alpha, t) = \frac{1+i\alpha}{1+\alpha^2} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1+i\alpha}{1+\alpha^2} e^{-k\alpha^2 t} e^{-i\alpha x} d\alpha.$$

Since

$$\frac{1+i\alpha}{1+\alpha^2} (\cos \alpha x - i \sin \alpha x) = \frac{\cos \alpha x + \alpha \sin \alpha x}{1+\alpha^2} + \frac{i(\alpha \cos \alpha x - \sin \alpha x)}{1+\alpha^2}$$

and the integral of the product of the second term with $e^{-k\alpha^2 t}$ is 0 (it is an odd function), we have

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\cos \alpha x + \alpha \sin \alpha x}{1+\alpha^2} e^{-k\alpha^2 t} d\alpha.$$

14. Using the Laplace transform the partial differential equation becomes

$$\frac{d^2 U}{dx^2} - sU = -100$$

so

$$U(x, s) = c_1 e^{-\sqrt{s} x} + c_2 e^{\sqrt{s} x} + \frac{100}{s}.$$

The condition $x \rightarrow \infty$ implies $\lim_{x \rightarrow \infty} U(x, s) = 100/s$ and the condition at $x = 0$ implies $U'(0, s) = -50/s$. thus $c_2 = 0$ and $c_1 = 50/s\sqrt{s}$, so

$$U(x, s) = \frac{100}{s} + 50 \frac{e^{-x\sqrt{s}}}{s\sqrt{s}}$$

and by (4) of Table 15.1 in the text,

$$u(x, t) = 100 + 100\sqrt{\frac{t}{\pi}} e^{-x^2/4t} - 50x \operatorname{erfc}\left(\frac{x}{2\sqrt{t}}\right).$$

15. Using the Fourier transform with respect to x we obtain

$$\frac{d^2 U}{dy^2} - \alpha^2 U = 0.$$

Since $0 < y < 1$ is a finite interval we use the general solution

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y.$$

The boundary condition at $y = 0$ transforms into $U'(\alpha, 0) = 0$, so $c_2 = 0$ and $U(\alpha, y) = c_1 \cosh \alpha y$. Now denote the Fourier transform of f as $F(\alpha)$. Then $U(\alpha, 1) = F(\alpha)$ so $F(\alpha) = c_1 \cosh \alpha$ and

$$U(\alpha, y) = F(\alpha) \frac{\cosh \alpha y}{\cosh \alpha}.$$

Taking the inverse Fourier transform we obtain

$$u(x, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) \frac{\cosh \alpha y}{\cosh \alpha} e^{-i\alpha x} d\alpha.$$

But

$$F(\alpha) = \int_{-\infty}^{\infty} f(t) e^{i\alpha t} dt,$$

and so

$$\begin{aligned} u(x, y) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(t) e^{i\alpha t} dt \right) \frac{\cosh \alpha y}{\cosh \alpha} e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{i\alpha(t-x)} \frac{\cosh \alpha y}{\cosh \alpha} dt d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) (\cos \alpha(t-x) + i \sin \alpha(t-x)) \frac{\cosh \alpha y}{\cosh \alpha} dt d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \alpha(t-x) \frac{\cosh \alpha y}{\cosh \alpha} dt d\alpha \\ &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \alpha(t-x) \frac{\cosh \alpha y}{\cosh \alpha} dt d\alpha, \end{aligned}$$

since the imaginary part of the integrand is an odd function of α followed by the fact that the remaining integrand is an even function of α .